

International Finance

Lecture VII – March 31st

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IES FSV UK – Spring 2026

Moodle: <https://dl1.cuni.cz/course/view.php?id=4605>

(International Finance JEB050/JPB335)

Course enrolment key: **PostUSD?2026**

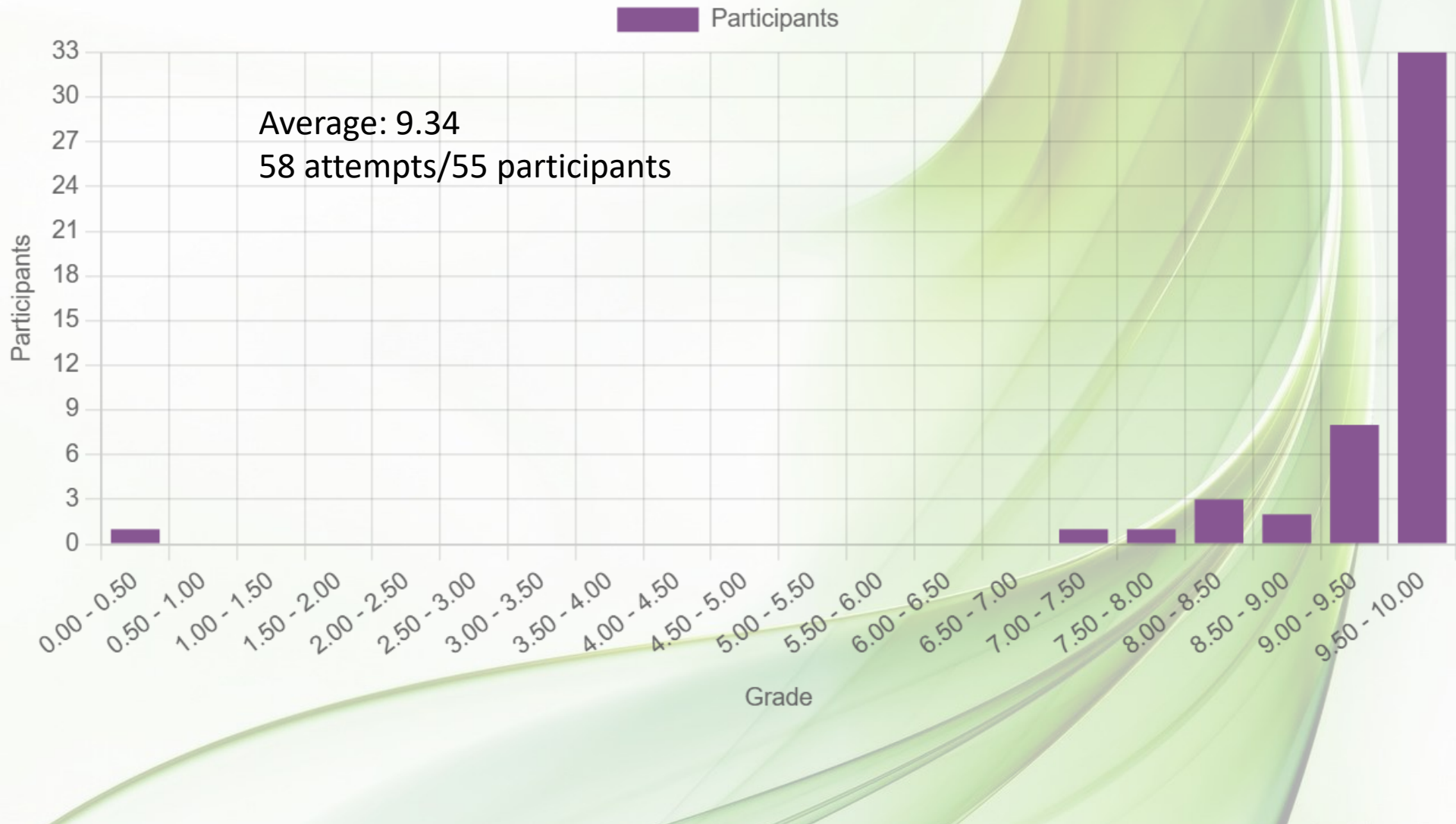
Course Outline (Changes Possible)

1. Introduction to international finance. Foreign exchange market: main actors, their role, types of interactions. Brief outline of the history of international finance. (February 17th)
2. International transfers/payments. Channels, related instruments. (February 24th)
3. Balance of payments structure and national income accounting for an open economy. International investment position (March 3rd)
4. Introduction to exchange rate determination: asset approach. Covered and uncovered interest rate parity. (March 10th)
5. Money, interest rates and exchange rates. Long-run aspects of exchange rate determination. Purchasing power parity. Overshooting. (March 17th)
6. Prices and exchange rates continued: Fischer effect. Exchange rates and competitiveness: real (effective) exchange rates, Balassa-Samuelson effect. **Application on the nominal and real convergence in the EU** (March 24th)
7. **Introduction to the AA-DD model. Temporary and permanent changes in fiscal and monetary policies.** (March 31st)
8. The AA-DD model: XX curve. Application on policy analysis. (April 7th)
9. Policy options for reaching internal and external balance. Swan diagram. Fixed exchange rates: macroeconomic implications. Interventions and sterilisation. Policy trilemma. (April 14th)
10. **Monetary integration: costs and benefits. Monetary integration: effects – empirical research.** (April 21st)
11. A brief introduction to the models of balance of payments crises. (April 28th)
12. Debts and (developing) countries. **Deadline for the team assignment/paper** (May 5th)
13. The future of IMS, role of the USD. Discussion of current topics. (May 12th)
14. **Reserved for a make-up class (possibly for April 21st). (May 19th)**

Outline

- Course organization
 - Attendance, quiz, assignments (Teams + topics)
 - Quiz #7: will be open by tomorrow (Wednesday) midnight
 - The usual deadline: next Monday midnight
- Back to Balassa-Samuelson and real exchange rate:
 - Real convergence in the EU (reading)
 - Effects of under/overvaluations
- AA-DD model: output and exchange rate interactions
 - DD curve
 - AA curve
 - Short-run (super) equilibrium

Quiz #6



Reading for the Next Week:

- Krugman & Obstfeld & Melitz: International Economics (chapter 17)
 - Appendix to chapter 17 explains the relationships between AA-DD and IS-LM models
- Mao Takongmo (2019): Revisiting the AA-DD Model in Zero Lower Bound
- Online resources:
 - International Economics (S. Suranovic) on AA-DD model
 - This online source, “International Economics: Theory and Policy” by Saylor Academy, can be quite useful too.
 - Specifically chapter 20: The AA-DD Model:
 - https://saylordotorg.github.io/text_international-economics-theory-and-policy/s23-the-aa-dd-model.html

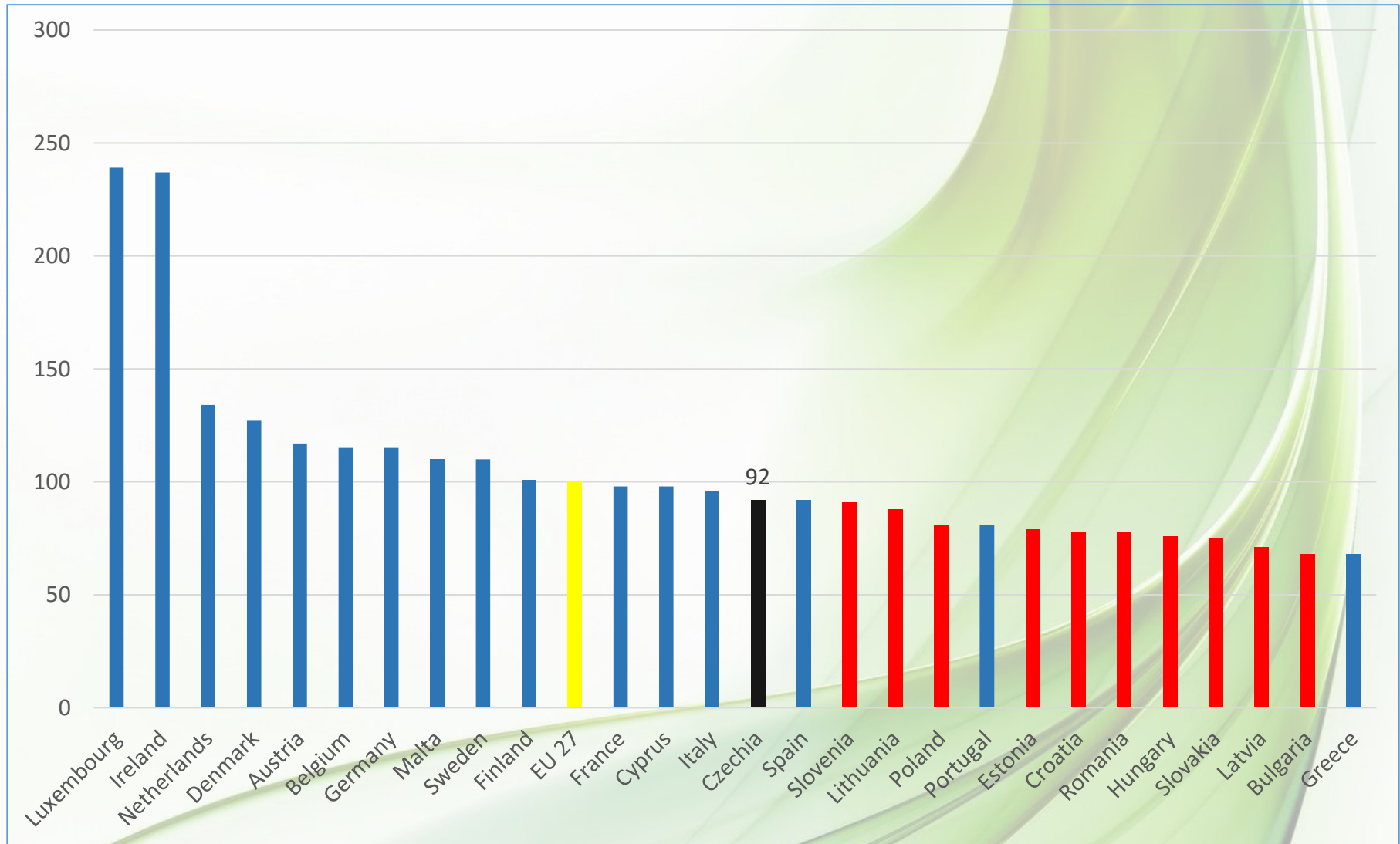
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Balassa-Samuelson and Discussion about Convergence in the EU?

- New data (released on March 25, 2026):
 - Eurostat - PPPs for GDP per capita in 2025: preliminary estimates
 - <https://ec.europa.eu/eurostat/web/products-eurostat-news/w/ddn-20260325-1>
 - <https://ec.europa.eu/eurostat/web/purchasing-power-parities/database>
 - Make sure that you understand the role of PPP in international comparisons of GDP per capita

Relative Level of Development

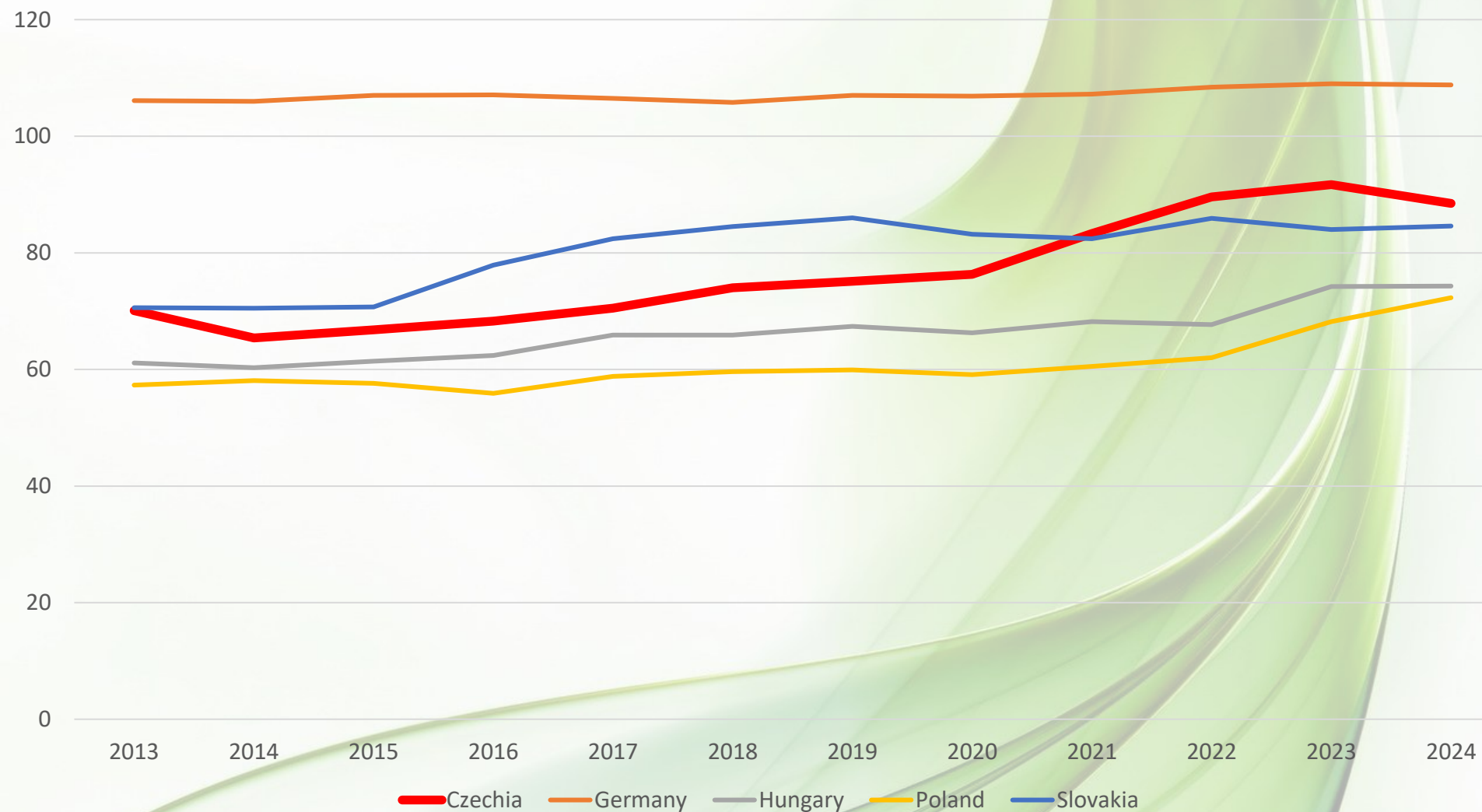
GDP per capita, PPS, 2025 (prel.) (%)



Source of data: Eurostat

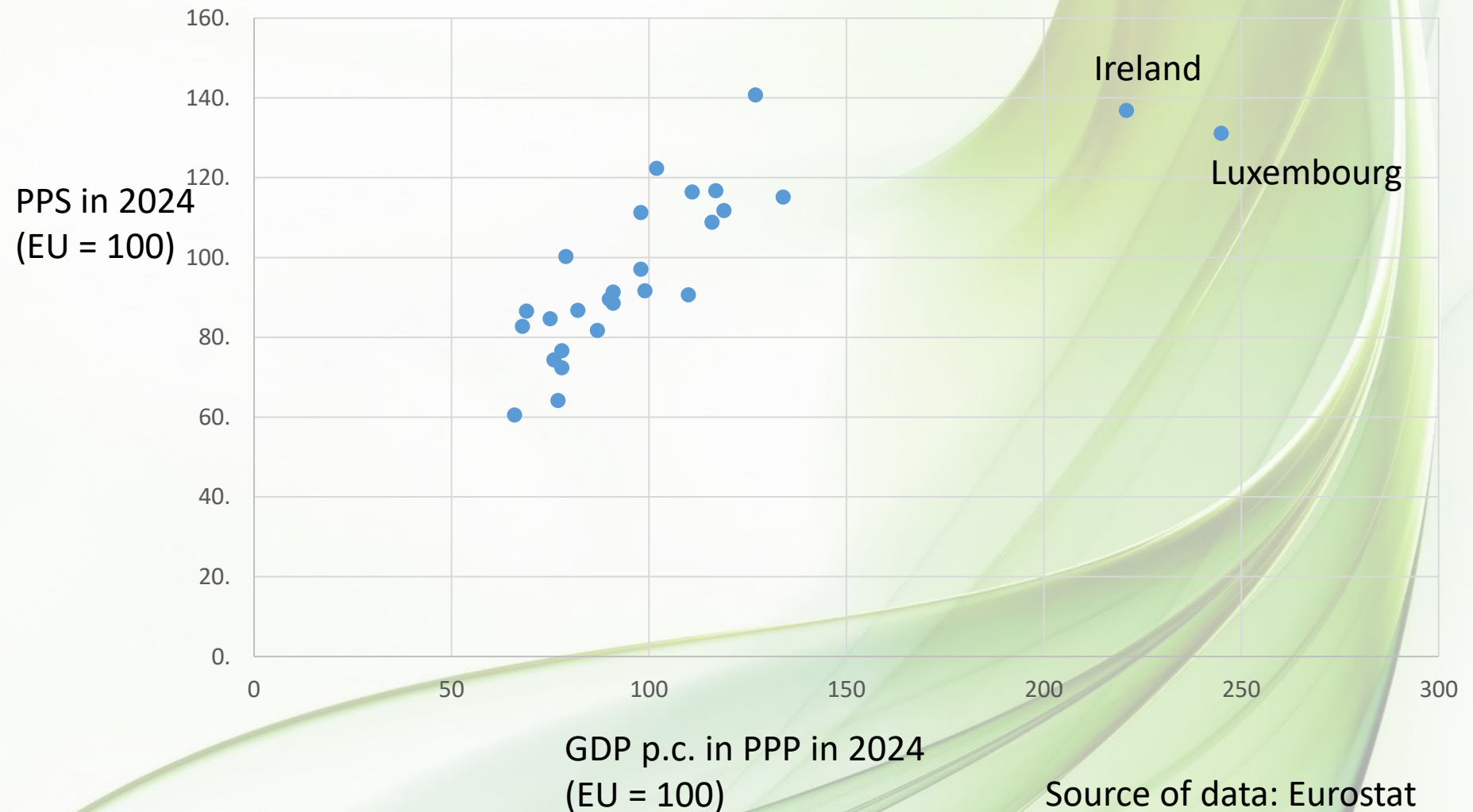
Price Level Indices for EU Countries

EU Average = 100

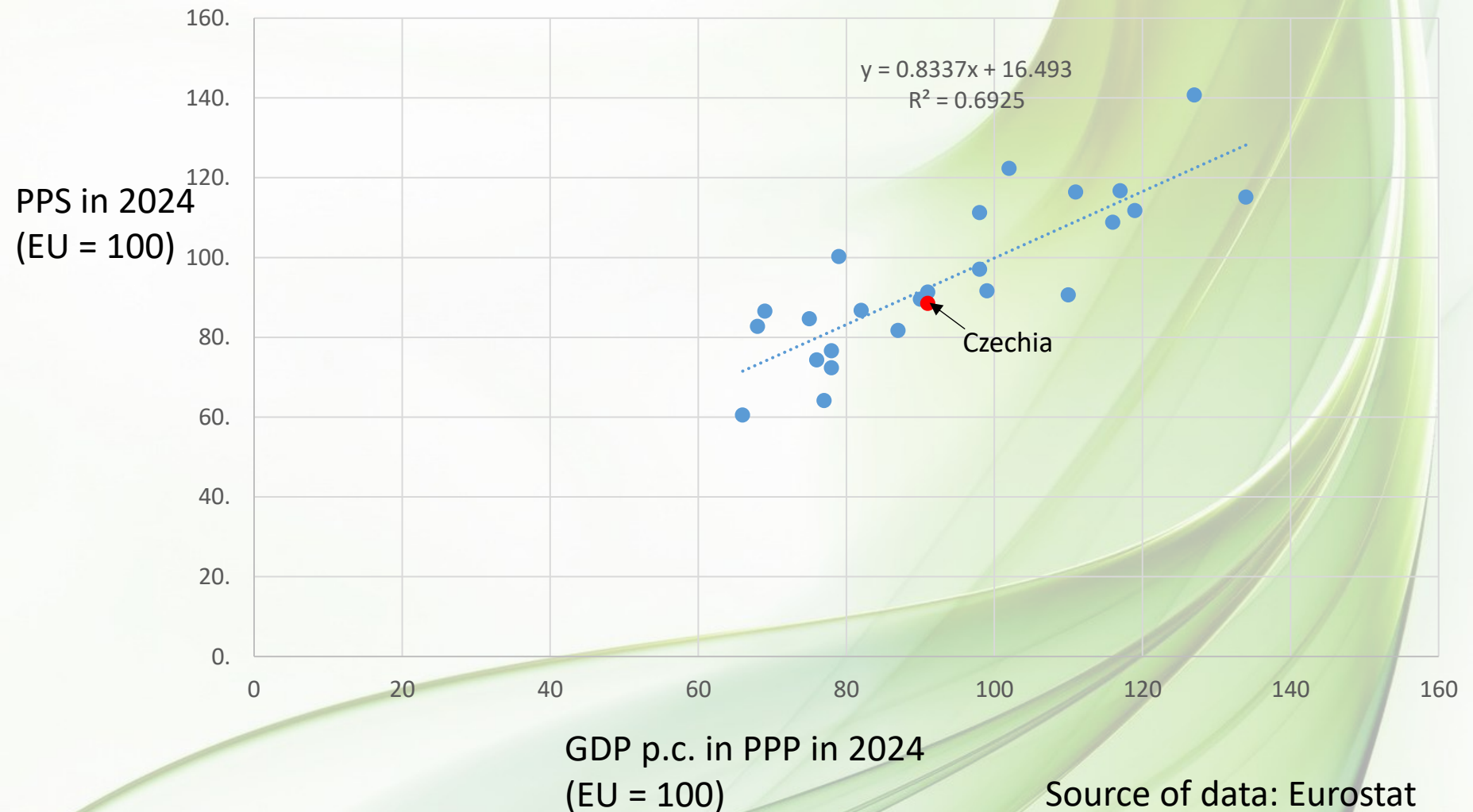


Source of data: Eurostat

GDP vs. Price Levels in 2024



GDP vs. Price Levels in 2024



Effects of Under/Overvaluations (Misalignments)



Debate about Effects of Misalignments (1)

- Baldwin & Krugman (1989): Persistent Trade Effects of Large Exchange Rate Shocks
 - This paper presents a theoretical basis for the argument that large exchange rate shocks may have **persistent effects on trade flows and the equilibrium exchange rate itself**.
 - *A model in which **large exchange rate fluctuations lead to entry or exit decisions that are not reversed** when the currency returns to its previous level.*
 - Hysteresis?
- Very interesting example of a rather significant shock (unfortunately “contaminated” with additional shocks):
 - East Germany in 1990/1991

Exchange Rates Before the Monetary Union

- Exchange rates between the Marks during the times of GDR
 - Official rate 1:1 (necessary for political reasons)
 - Internal GDR rate for trade 4.4:1
 - Black market exchange rates estimated to be between 5:1 – 15:1
- Other former CPEs performed substantial initial devaluation that brought their exchange rates towards convertible currencies close to the original black market levels
- What did the GDR do?

Helmut Kohl in 1990

- Slogan for Parliamentary election:
 - *„Nobody will be worse off and many will be better off“.*

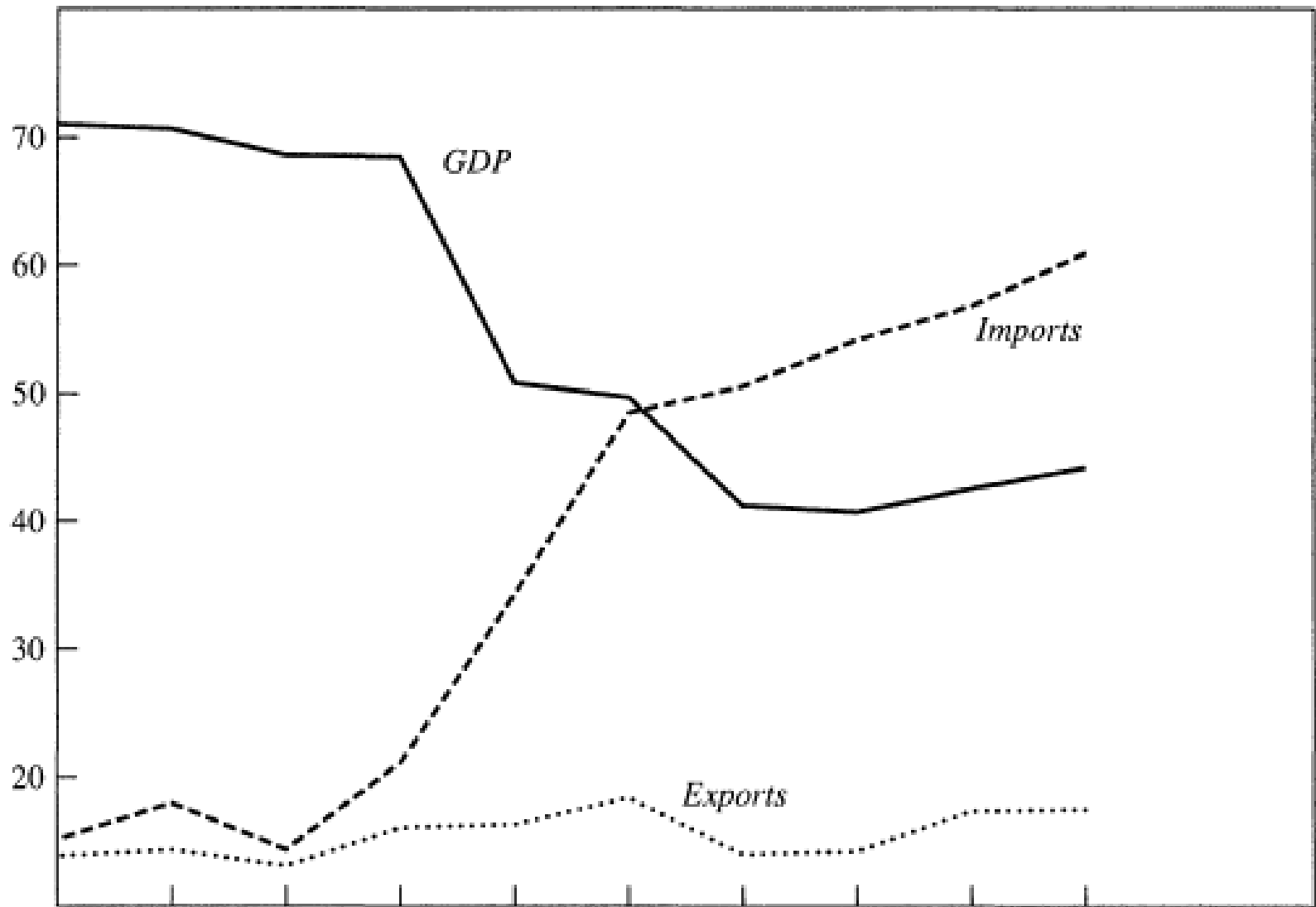
Conversion Rates

- For flows: 1 DEM per 1 DDM
- For assets:
 - German citizens – part of assets at 1:1, the rest at 2:1
 - 25 years and younger: 2000 DM at 1 DEM per 1 DM
 - 26 – 60 years old: 4000 DM
 - Above 60: 6000 DM
 - Assets held by non-residents accrued after 1989: 1 DEM per 3 DDM
 - Other assets: 1 DEM per 2 DDM (including e.g. debts of firms)

Implications: conversion rates made the price of East German labor too high (relative to productivity) and debts of the firms too big

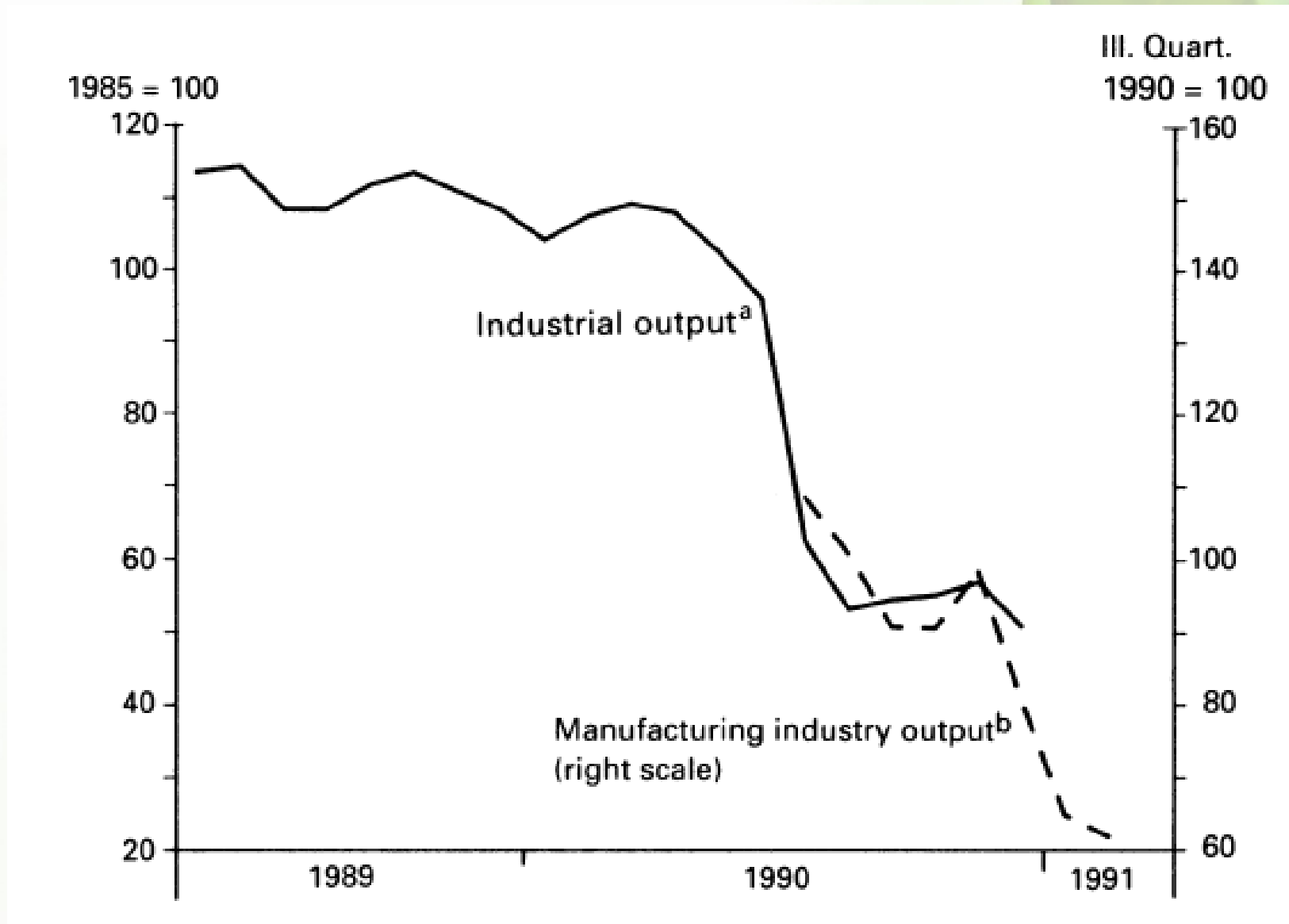
Trade Shock for Former GDR

Billions of 1990: 2H deutsche marks, quarterly rated



Source: Dornbusch, Wold, Alexander (1992)

East-German Industry



Source: Siebert (1991)

Employment

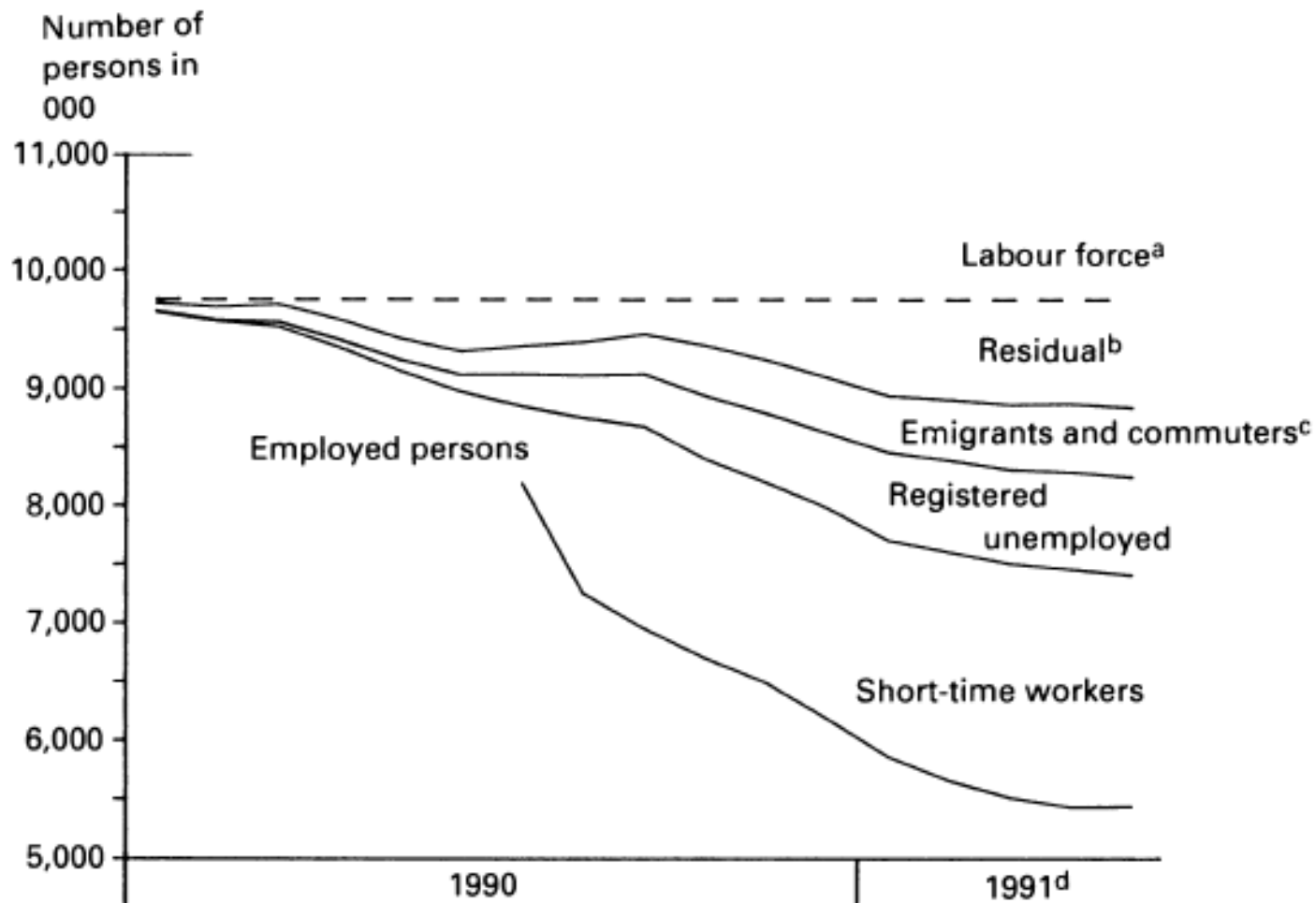
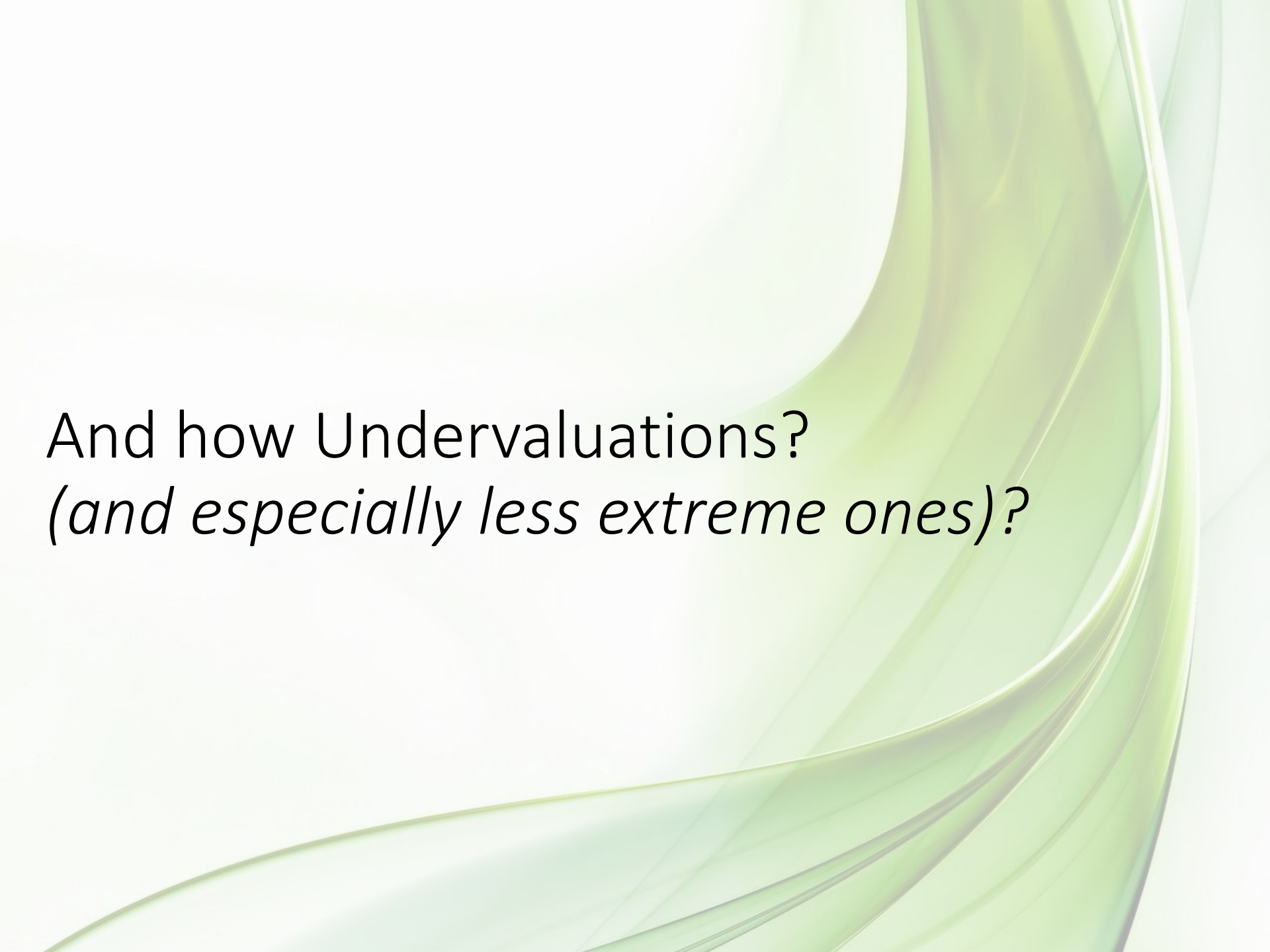


Figure 4. The U-curve of employment in Eastern Germany

Source: Own calculations.

Notes: (a) IV. quarter 1989; (b) E.g. persons in early retirement, unregistered persons; (c) Partly estimated on a quarterly basis; (d) Partly estimated.

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And how Undervaluations?
(and especially less extreme ones)?

Debate about Effects of Misalignments (2)

- Rodrik (2008) tests effects of undervaluation (misalignments) for growth?
- Gist of the approach:
 - Assume that real GDP predicts equilibrium real exchange rate

$$\ln RER_{it} = \alpha + \beta \ln RGDPC H_{it} + f_i + u_{it}$$

- Calculate misalignment: $\ln UNDERVAL_{it} = \ln RER_{it} - \ln \widehat{RER}_{it}$
- Test whether misalignment correlates with growth
$$growth_{it} = \alpha + \beta \ln RGDPC H_{i,t-1} + \delta \ln UNDERVAL_{it} + f_i + f_t + u_{it}$$
- Conclusion?
 - Undervaluation good for growth?

Rodrik (2008) Regression Results

Table 1. Baseline Panel Regressions of Economic Growth on the Undervaluation Measure^a

Independent variable	Sample					
	<i>All countries, 1950–2004</i> 1-1	<i>Developed countries,^b 1950–2004</i> 1-2	<i>Developing countries, 1950–2004</i> 1-3	<i>All countries, 1950–2004</i> 1-4	<i>Developing countries, 1950–79</i> 1-5	<i>Developing countries, 1980–2004</i> 1-6
ln initial income	–0.031*** (–6.67)	–0.055*** (–6.91)	–0.039*** (–5.30)	–0.032*** (–7.09)	–0.062*** (–3.90)	–0.065*** (–4.64)
ln <i>UNDerval</i>	0.017*** (5.21)	0.003 (0.49)	0.026*** (5.84)	0.086*** (4.05)	0.029*** (4.20)	0.024*** (3.23)
ln initial income × ln <i>UNDerval</i>				–0.0087*** (–3.39)		
No. of observations	1,303	513	790	1,303	321	469

Source: Author's regressions.

a. The dependent variable is annual growth in GDP per capita, in percent. Observations are five-year averages. All regressions include time and country fixed effects. Countries with extreme observations for *UNDerval* (Iraq, Laos, and North Korea) have been excluded from the samples. Robust *t* statistics are in parentheses. Asterisks indicate statistical significance at the *10 percent, **5 percent, or ***1 percent level.

b. Developed country observations are those with real GDP per capita exceeding \$6,000.

Problems with the conclusions?

How About CEE Countries?

- Cuestas et al (2020): Real exchange rate misalignments in CEECs: Have they hindered growth?
 - Exchange rate **overvaluation** has a **negative impact** on economic activity.
 - It is harmful for economic activity if the exchange rate is overvalued as was found in most of the previous literature such as Berg and Miao (2010), MacDonald and Vieira (2010), Schröder (2013), Comunale (2017) and Habib et al. (2017).
 - The effect of misalignments on economic activity seems to be nonlinear, as **overvaluation** has a **stronger effect than undervaluation**.
 - This asymmetric effect is in line with previous results from Aguirre and Calderón (2005) for a sample of 60 countries.
 - **Undervaluation** has had a **limited effect on economic activity** in the experience of CEE countries.

Exchange rate and macroeconomics in the short run.

DD-AA Model

This part is based on official slides for the Krugman-Obstfeld-Melitz book.

DD-AA Model

- A model which provides analysis of three markets in a simple framework
 - **Output** – markets of goods and services
 - Foreign exchange markets
 - Money market
- Short run focus
- Two curves:
 - **DD**: Shows combinations of **output** and the **exchange rate** at which the output market is in short-run equilibrium
 - **AA**: The inverse relationship between **output** and **exchange rates** needed to keep the foreign exchange markets and the money market in equilibrium

DD Curve

Combinations of output and the exchange rate at which the output market is in short-run equilibrium

Determinants of Aggregate Demand (cont.)

- Aggregate demand can be expressed as:

$$D = C + I + G + CA \quad (\text{Note: } CA \equiv X - M)$$

$$D = C \overset{(+)}{(\mathbf{Y} - \mathbf{T})} + \overset{(+)}{\mathbf{I} + \mathbf{G}} + CA \overset{(+)}{(\mathbf{EP}^* / \mathbf{P}, \overset{(-)}{\mathbf{Y} - \mathbf{T}})}$$

Or more simply: $D = D \overset{(+)}{(\mathbf{EP}^* / \mathbf{P}, \overset{(+)}{\mathbf{Y} - \mathbf{T}}, \overset{(+)}{\mathbf{I}}, \overset{(+)}{\mathbf{G}})$

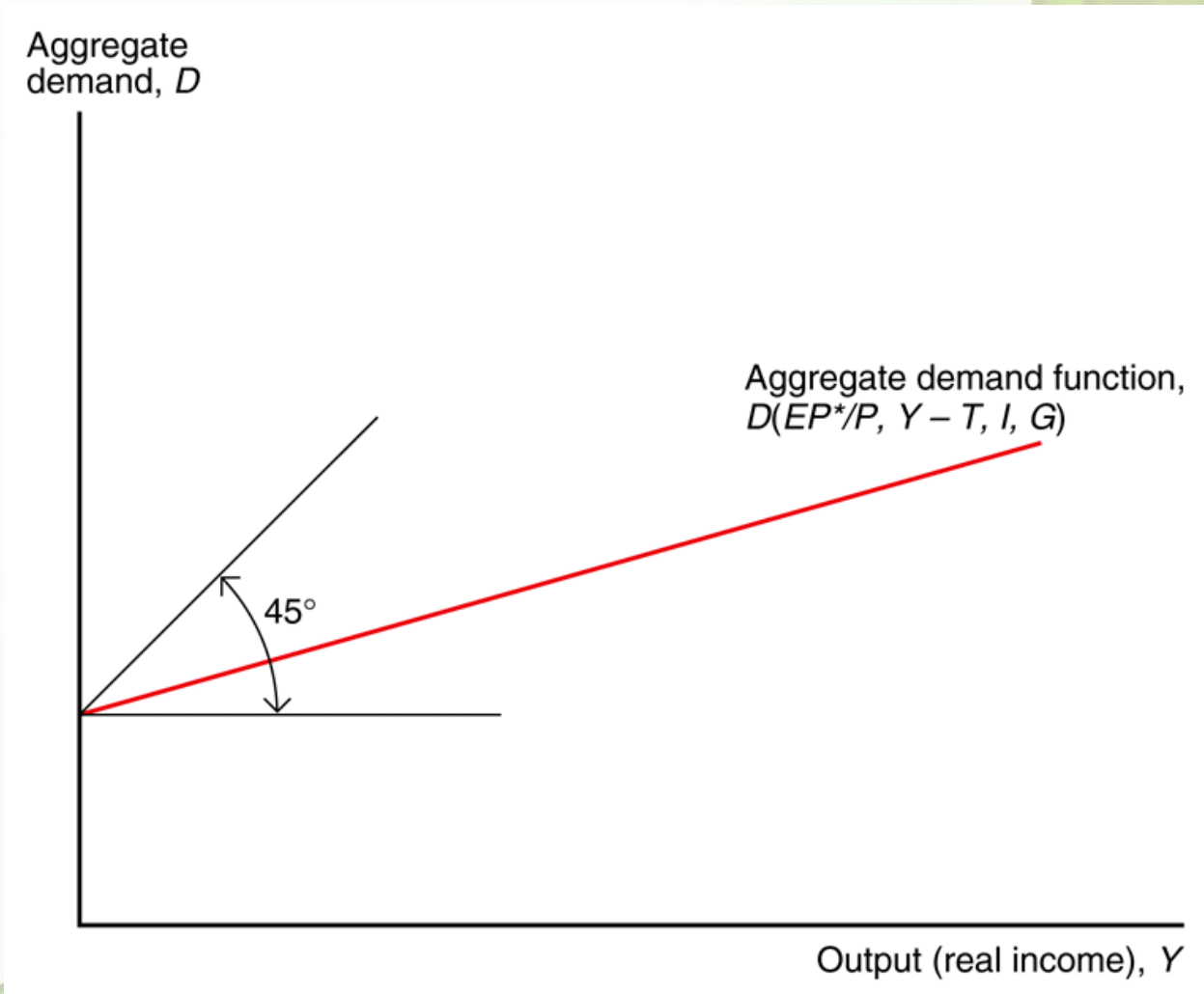
Determinants of Aggregate Demand (cont.)

- Determinants of aggregate demand include:
 - **Real exchange rate:** an increase in the real exchange rate increases the current account, and therefore increases aggregate demand of domestic products.
 - **Disposable income:** an increase in the disposable income increases consumption expenditure, but decreases the current account.
 - Since consumption expenditure is usually greater than expenditure on foreign products, the first effect dominates the second effect.
 - As income increases for a given level of taxes, aggregate consumption expenditure and aggregate demand increase by less than income.

Determinants of Aggregate Demand

- For simplicity, we assume that **exogenous** political factors determine government purchases **G** and the level of taxes **T** .
- For simplicity, we currently assume that investment expenditure **I** is determined by **exogenous** business decisions.
 - A more complicated model shows that investment depends on the cost of spending or borrowing to finance investment: the interest rate.

Diagram: Aggregate Demand as a Function of Output

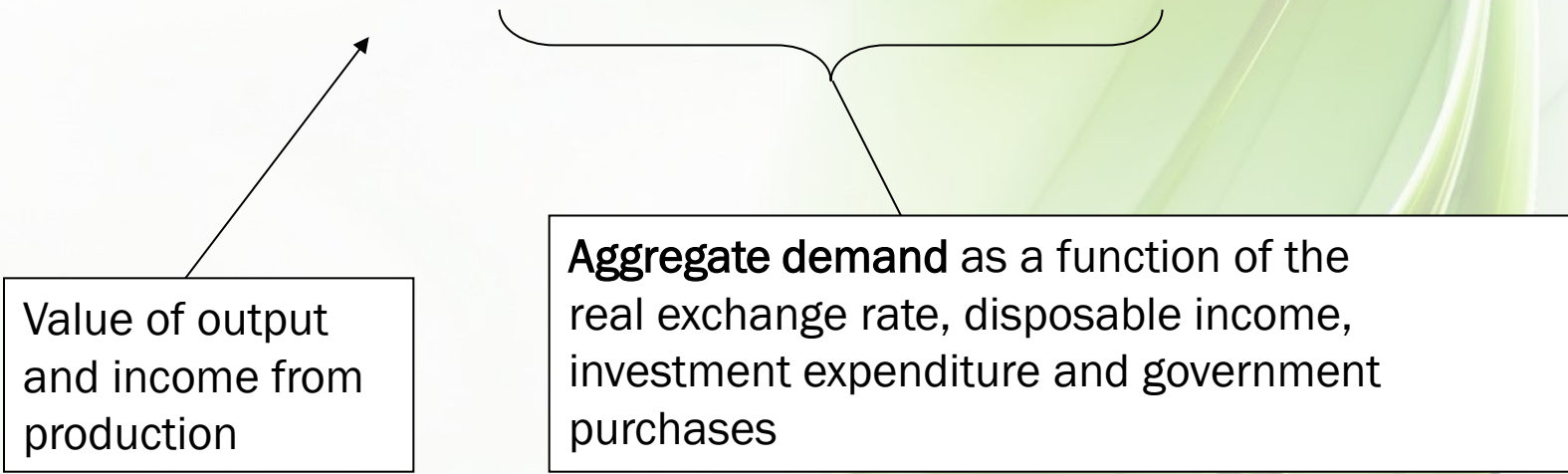


Short-Run Equilibrium for Aggregate Demand and Output

- Equilibrium is achieved when the value of income from production (output) Y equals the value of aggregate demand D .

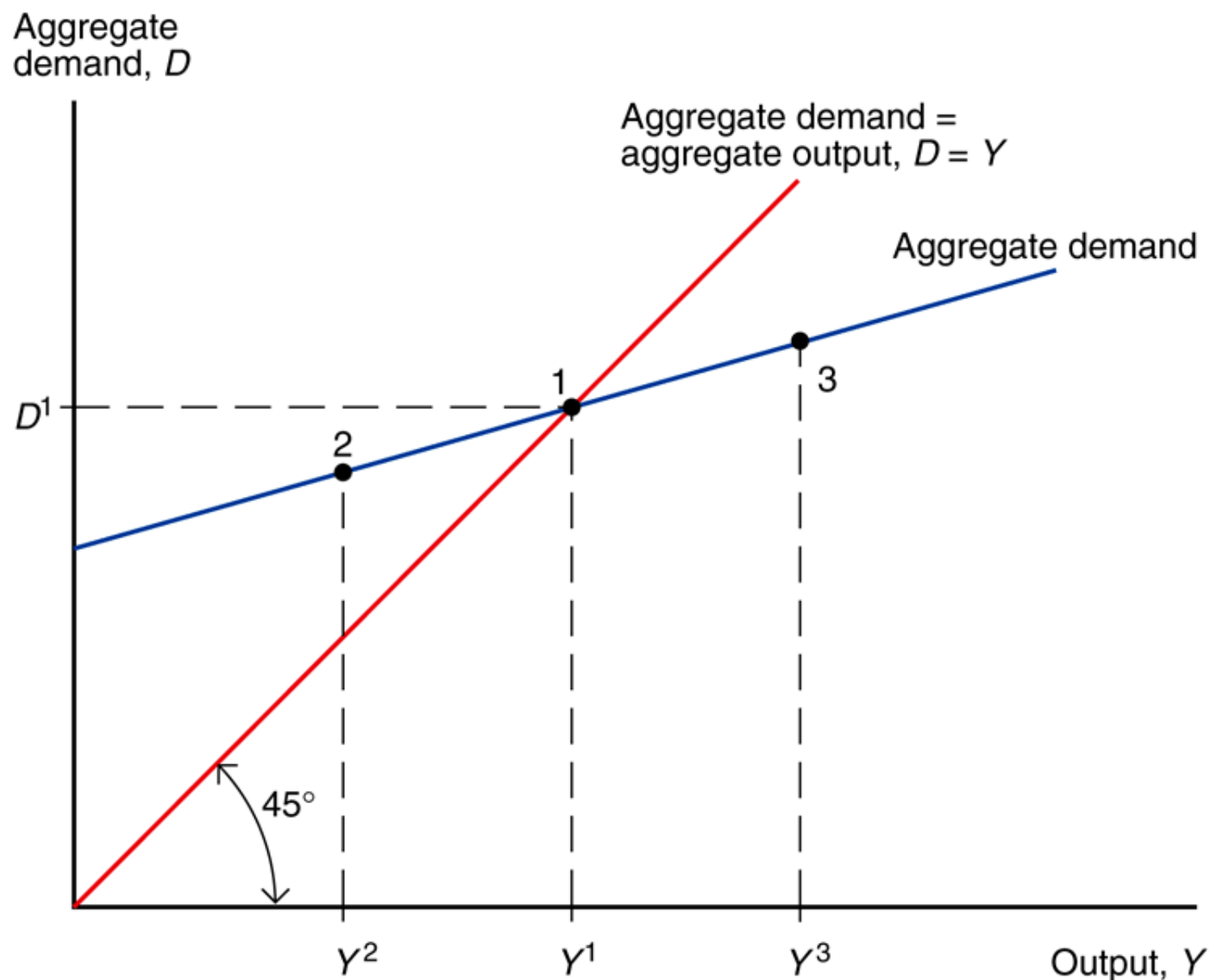
$$Y = D(EP^*/P, Y - T, I, G)$$

Value of output
and income from
production



Aggregate demand as a function of the
real exchange rate, disposable income,
investment expenditure and government
purchases

The Determination of Output in the Short Run



Short-Run Equilibrium and the Exchange Rate:

DD Schedule

- With fixed domestic and foreign levels of average prices, a **rise in the nominal exchange rate makes foreign goods and services more expensive** relative to domestic goods and services.
- A rise in the nominal exchange rate (a domestic currency depreciation) therefore **increases aggregate demand of domestic products**.
- In equilibrium, **production will increase to match the higher aggregate demand**.

Output Effect of a Currency Depreciation with Fixed Output Prices

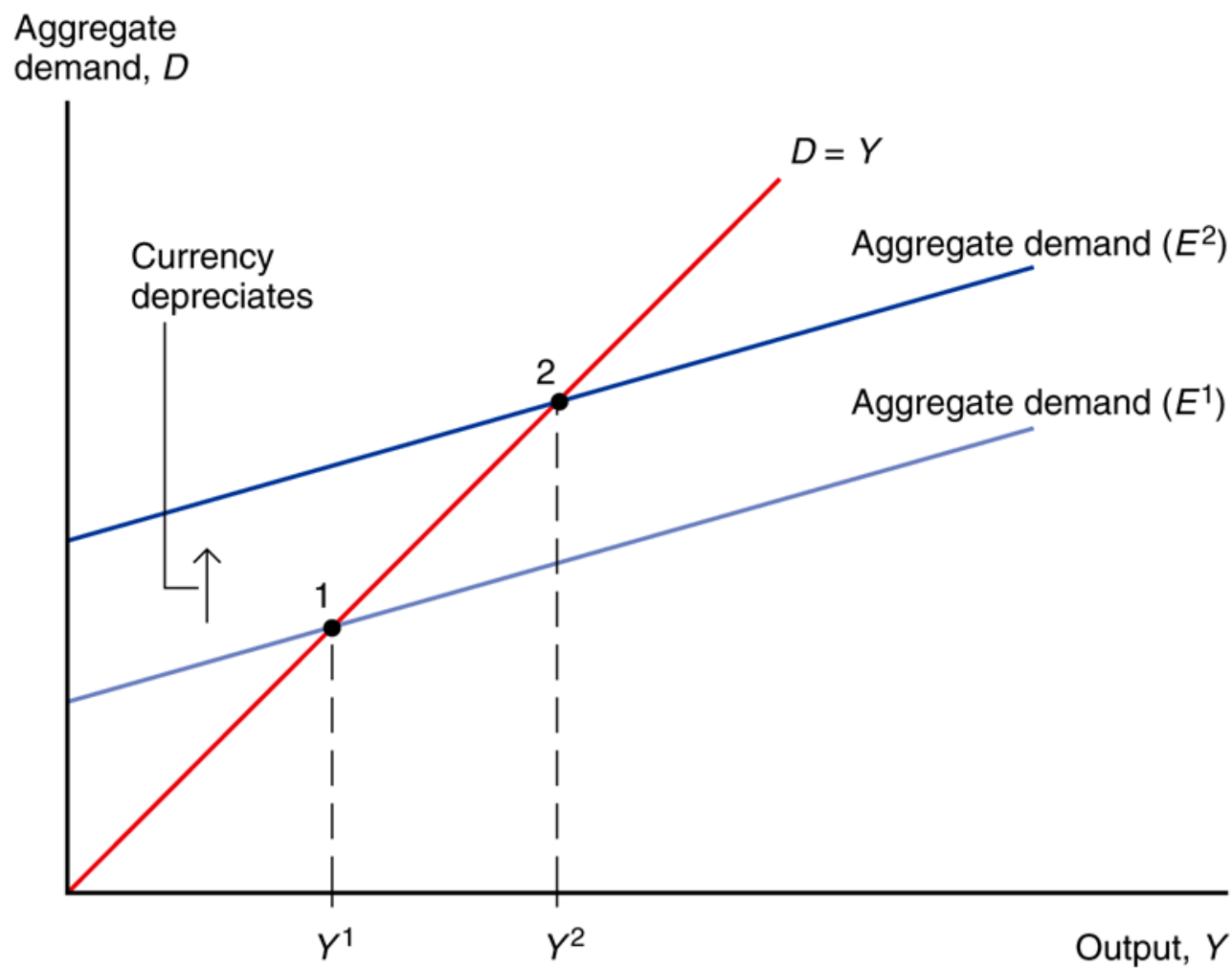
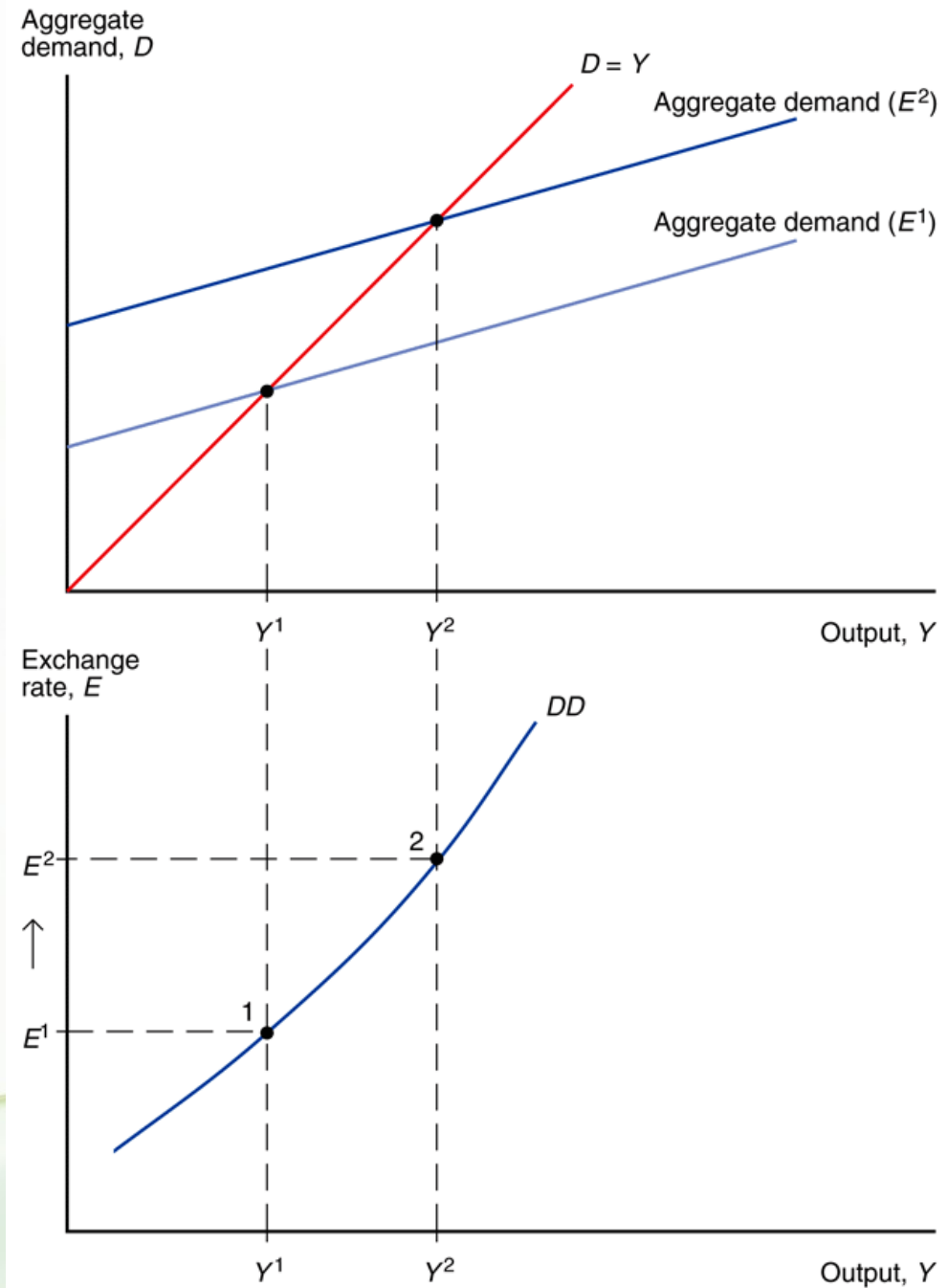


Fig. 17-4:
Deriving the
DD Schedule



Short-Run Equilibrium and the Exchange Rate: *DD* Schedule

***DD* schedule**

- Shows combinations of output and the exchange rate at which the output market is in short-run equilibrium (such that aggregate demand = aggregate output).
- **Slopes upward because a rise in the exchange rate causes aggregate demand and aggregate output to rise.**

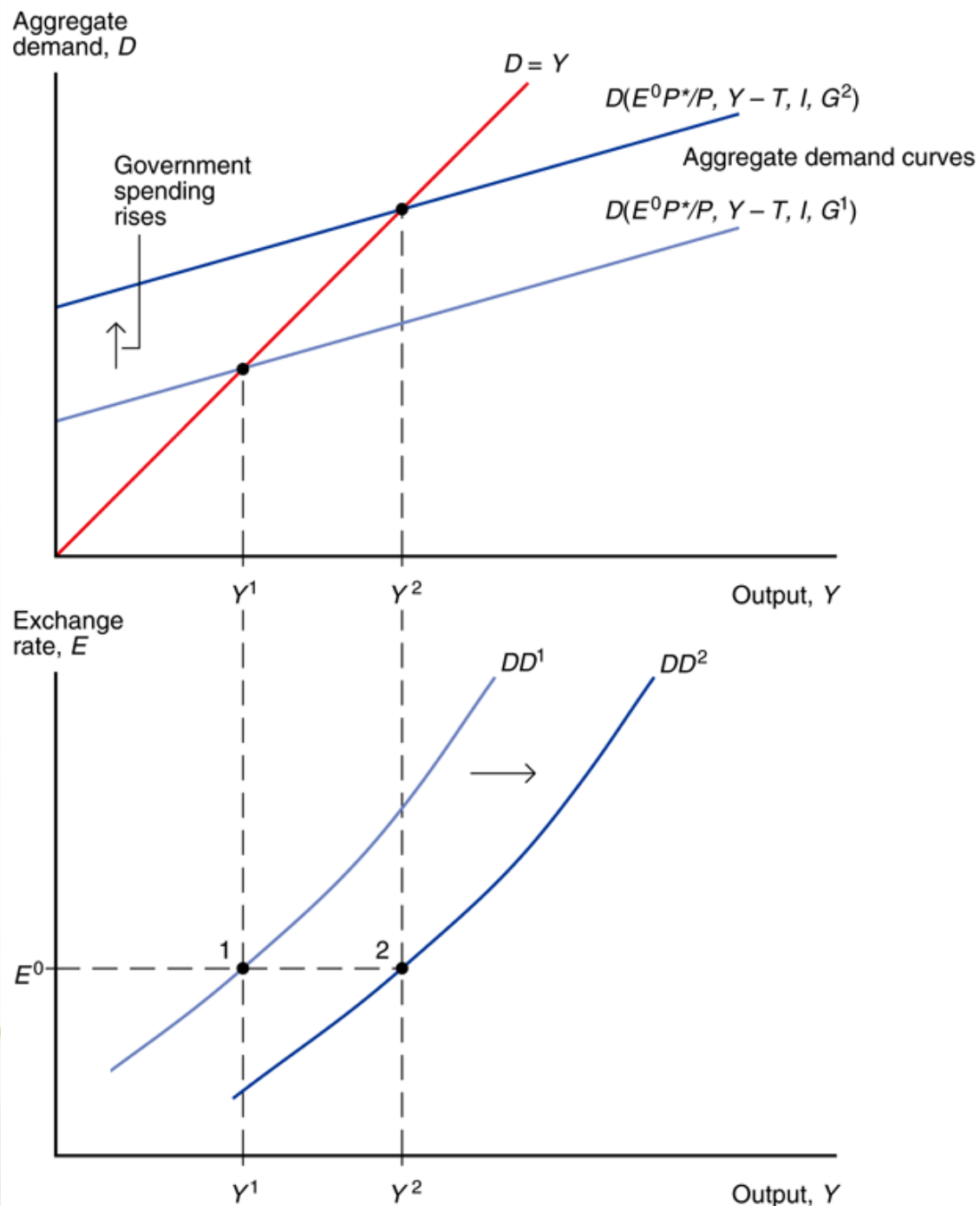
Shifting the *DD* Curve

Changes in the exchange rate cause movements along a *DD* curve.

Other changes cause it to **shift**:

1. **Changes in *G*: higher government purchases** cause higher aggregate demand and output in equilibrium. Output increases for every exchange rate: **the *DD* curve shifts right.**

Fig. 17-5:
Government
Demand and the
Position of the *DD*
Schedule



Shifting the DD Curve (2)

2. **Changes in T : lower taxes** generally increase consumption expenditure, increasing aggregate demand and output in equilibrium for every exchange rate: the **DD curve shifts right**.
3. **Changes in I : higher investment** expenditure is represented by shifting the **DD curve right**.
4. **Changes in P relative to P^* : lower domestic prices** relative to foreign prices are represented by shifting the **DD curve right**.

Shifting the *DD* Curve (3)

5. **Changes in *C*:** willingness to consume more and save less is represented by **shifting the *DD* curve right**.
6. **Changes in demand of domestic goods relative to foreign goods:** willingness to consume more domestic goods relative to foreign goods is represented by **shifting the *DD* curve right**.

AA Curve

Equilibrium in asset markets

Short-Run Equilibrium in Asset Markets

We consider two sets of asset markets:

1. Foreign exchange markets

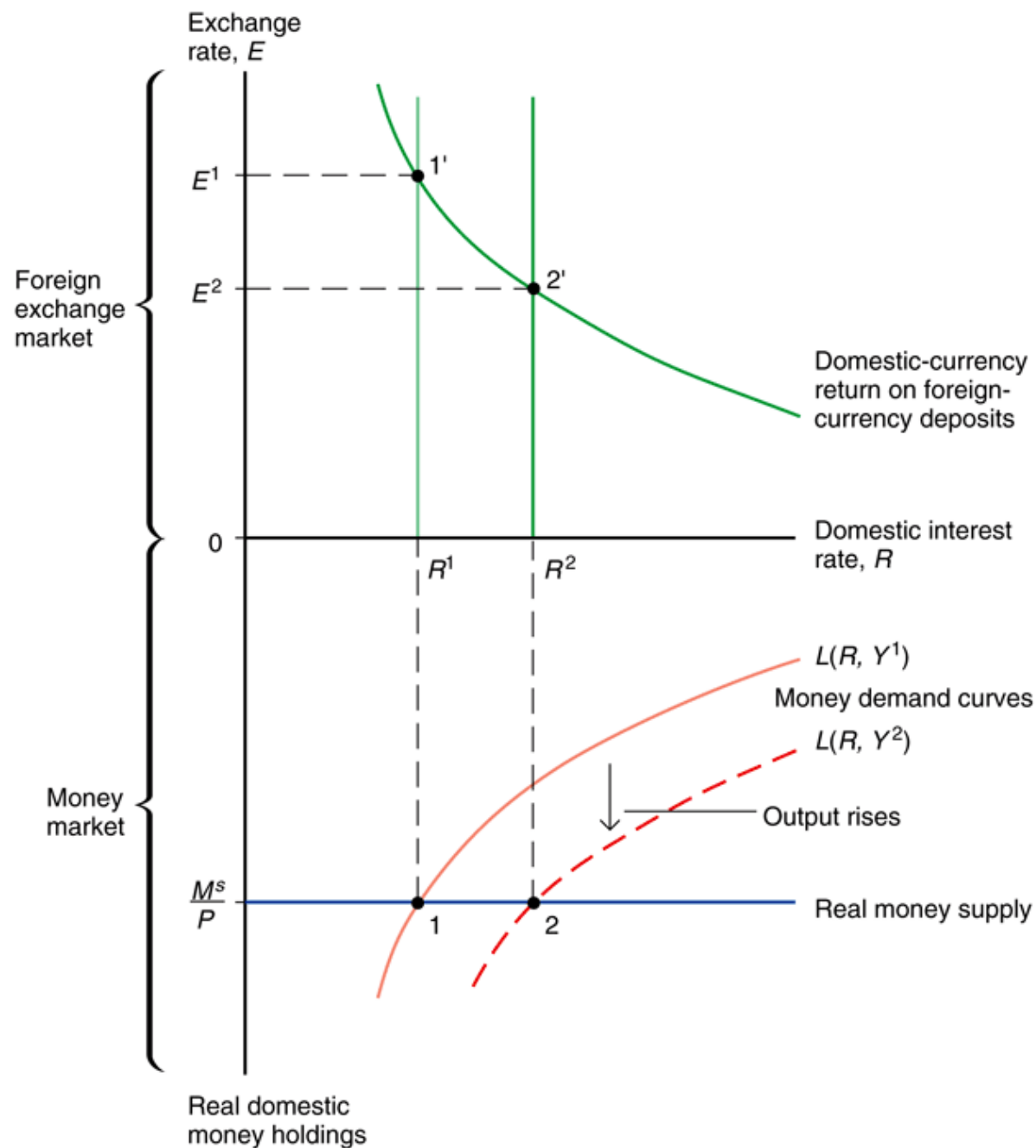
- interest parity represents equilibrium:

$$R = R^* + (E^e - E)/E$$

2. Money market

- Equilibrium occurs when the quantity of real monetary assets supplied matches the quantity of real monetary assets demanded: $M^s/P = L(R, Y)$
- A rise in income from production causes the demand of real monetary assets to increase.

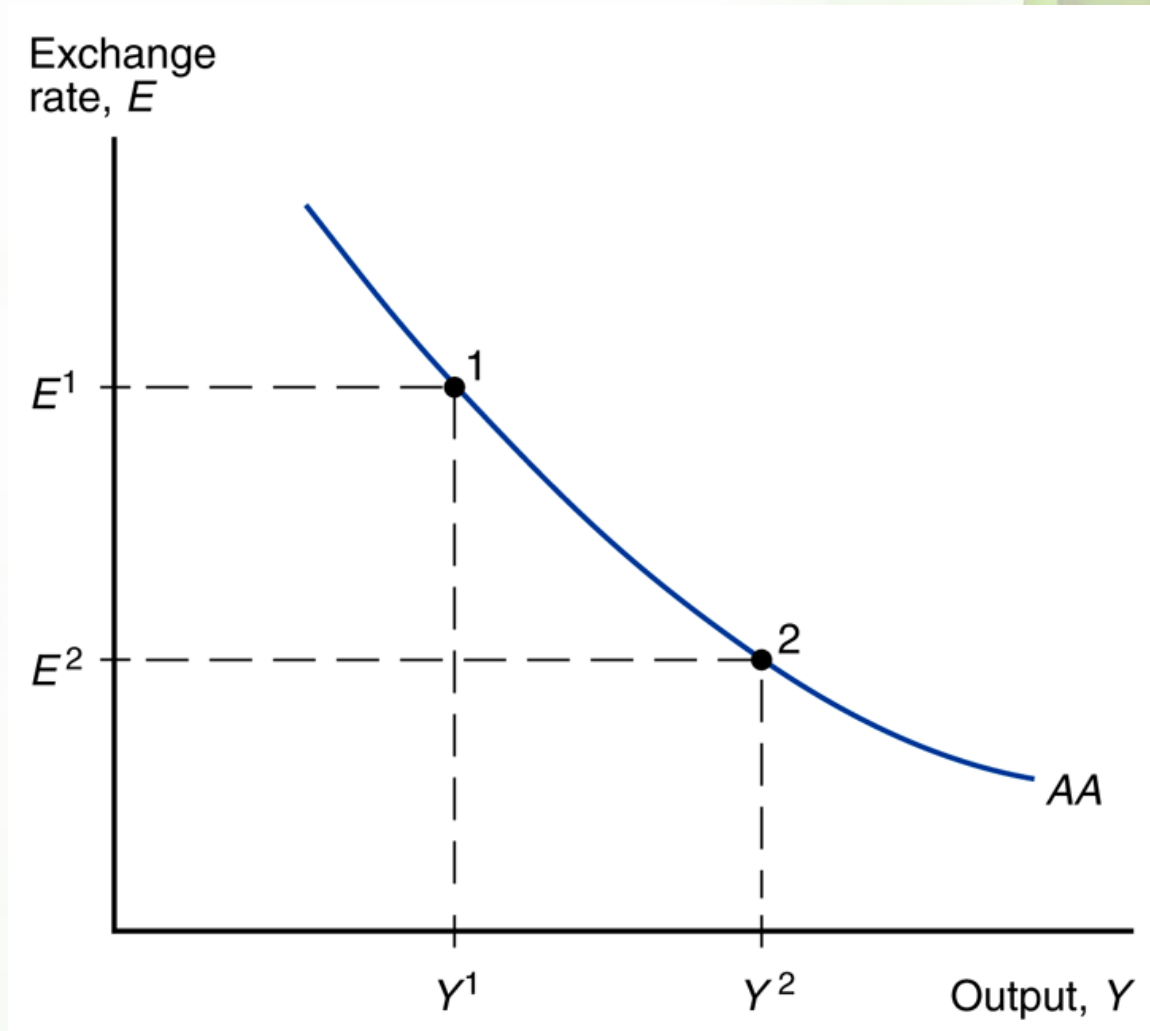
Output and the Exchange Rate in Asset Market Equilibrium



Short-Run Equilibrium in Asset Markets (cont.)

- When **income and production increase**,
 - demand for real monetary assets increases,
 - an increase in domestic interest rates,
 - leading to an **appreciation of the domestic currency**.
- *Recall that an appreciation of the domestic currency = fall in E .*
- When income and production decrease, the domestic currency depreciates and E rises.

The AA Schedule



AA curve - the inverse relationship between output and exchange rates needed to keep the foreign exchange markets and the money market in equilibrium

Shifting the AA Curve

1. **Changes in M^s :** an **increase in the money** supply reduces interest rates in the short run, causing the domestic currency to depreciate (a rise in E) for every Y : **the AA curve shifts up** (right).

Shifting the AA Curve (cont.)

2. **Changes in P :** An **increase in the level of average domestic prices** decreases the supply of real monetary assets, increasing interest rates, causing the domestic currency to appreciate (a fall in E): the **AA curve shifts down** (left).
3. **Changes in the demand of real monetary assets:** if domestic residents are willing to hold a lower amount of real money assets and more non-monetary assets, interest rates on nonmonetary assets would fall, leading to a depreciation of the domestic currency (a rise in E): the **AA curve shifts up** (right).

Shifting the AA Curve (cont.)

4. **Changes in R^* :** An **increase in the foreign interest rates** makes foreign currency deposits more attractive, leading to a depreciation of the domestic currency (a rise in E): **the AA curve shifts up (right).**
5. **Changes in E^e :** if market participants ***expect the domestic currency to depreciate in the future***, foreign currency deposits become more attractive, causing the domestic currency to depreciate (a rise in E): ***the AA curve shifts up (right).***

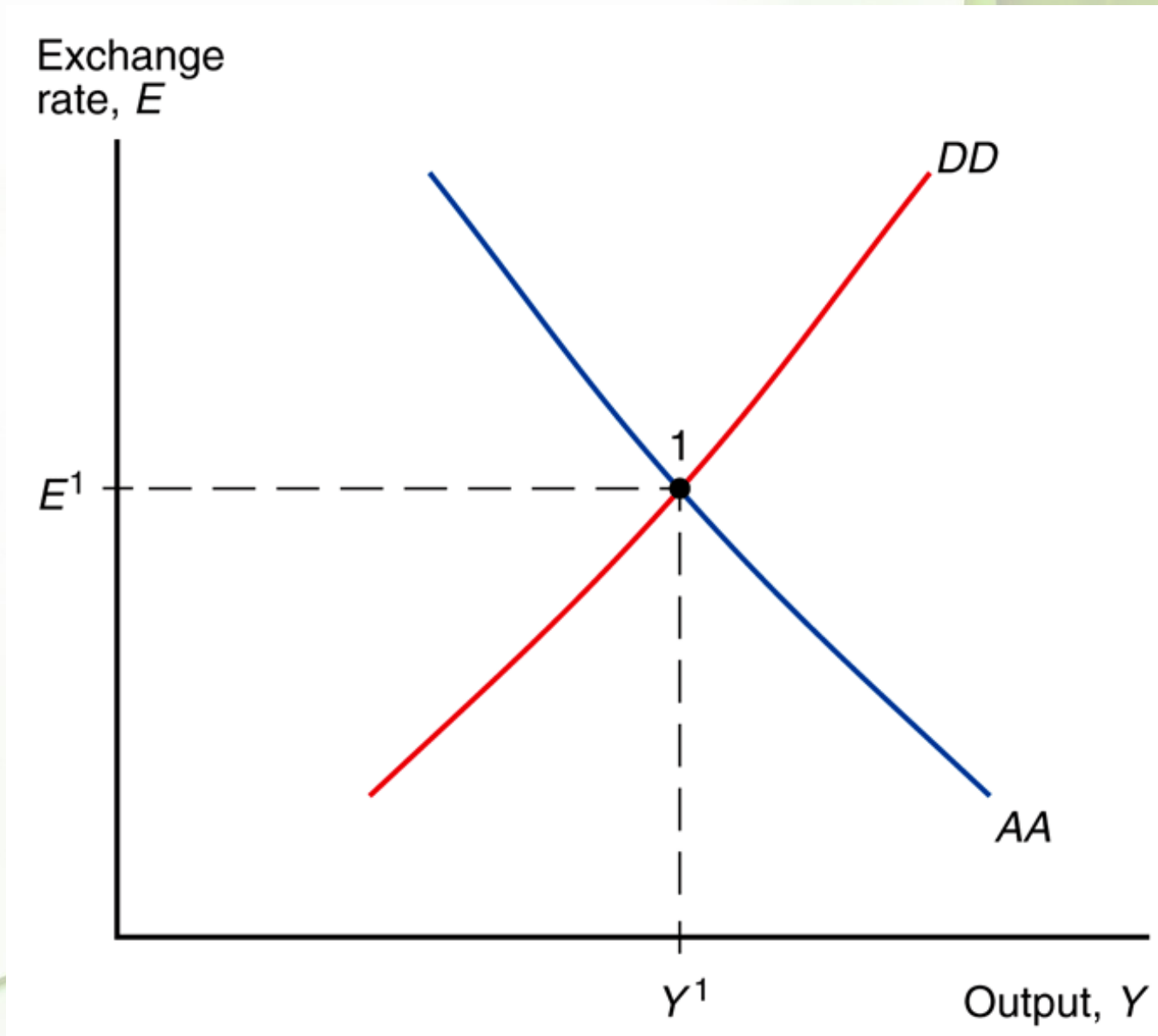
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DD-AA: SR Equilibrium

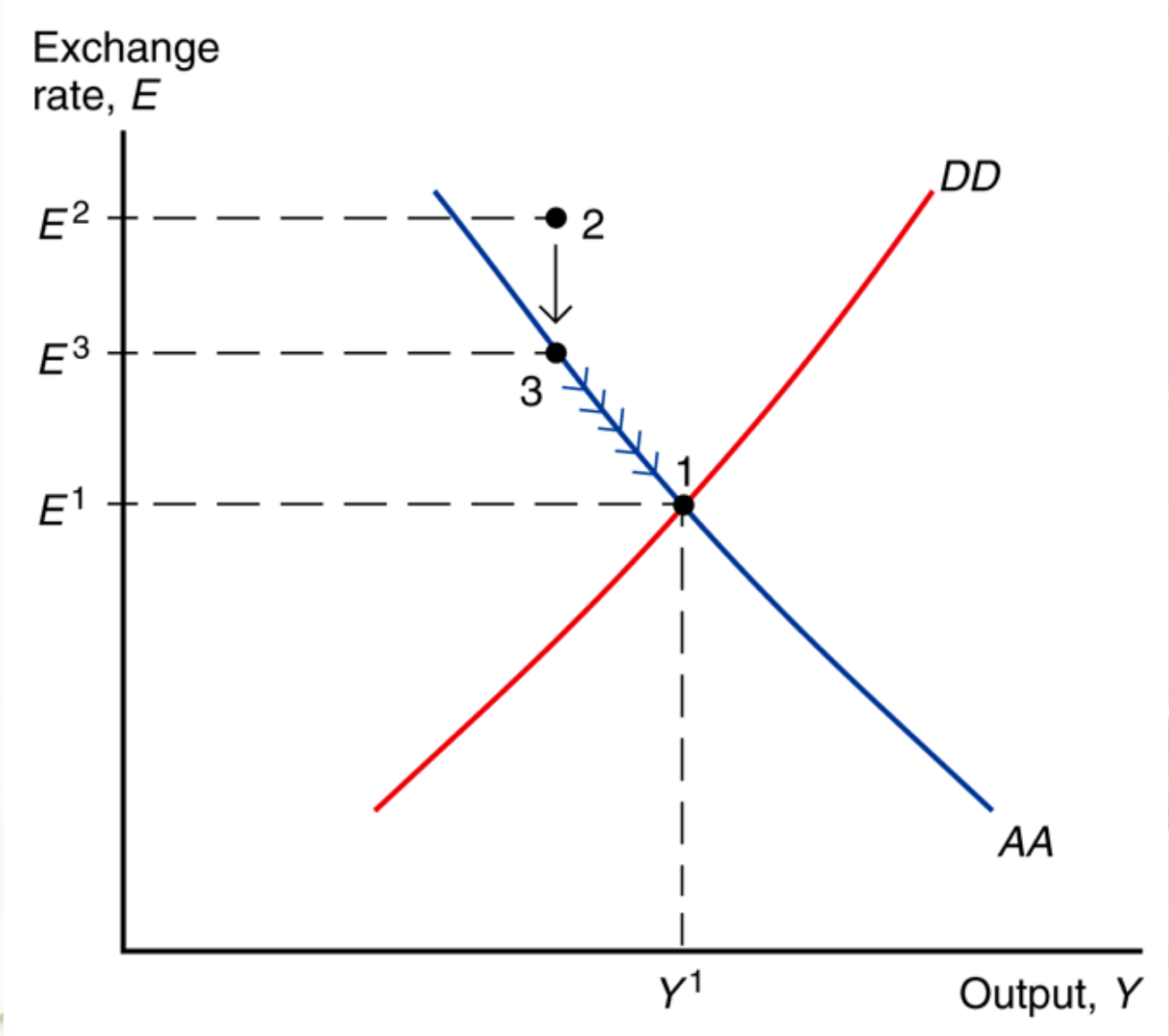
Putting the Pieces Together: the *DD* and *AA* Curves

- A **short-run equilibrium** means a *nominal exchange rate* and level of *output* such that
 1. equilibrium in the output markets holds: aggregate demand equals aggregate output.
 2. equilibrium in the foreign exchange markets holds: interest parity holds.
 3. equilibrium in the money market holds: the quantity of real monetary assets supplied equals the quantity of real monetary assets demanded.
- A **short-run equilibrium** occurs at the intersection of the *DD* and *AA* curves:
 - output markets are in equilibrium on the *DD* curve
 - asset markets are in equilibrium on the *AA* curve

Short-Run Equilibrium: The Intersection of DD and AA



How the Economy Reaches Its Short-Run Equilibrium

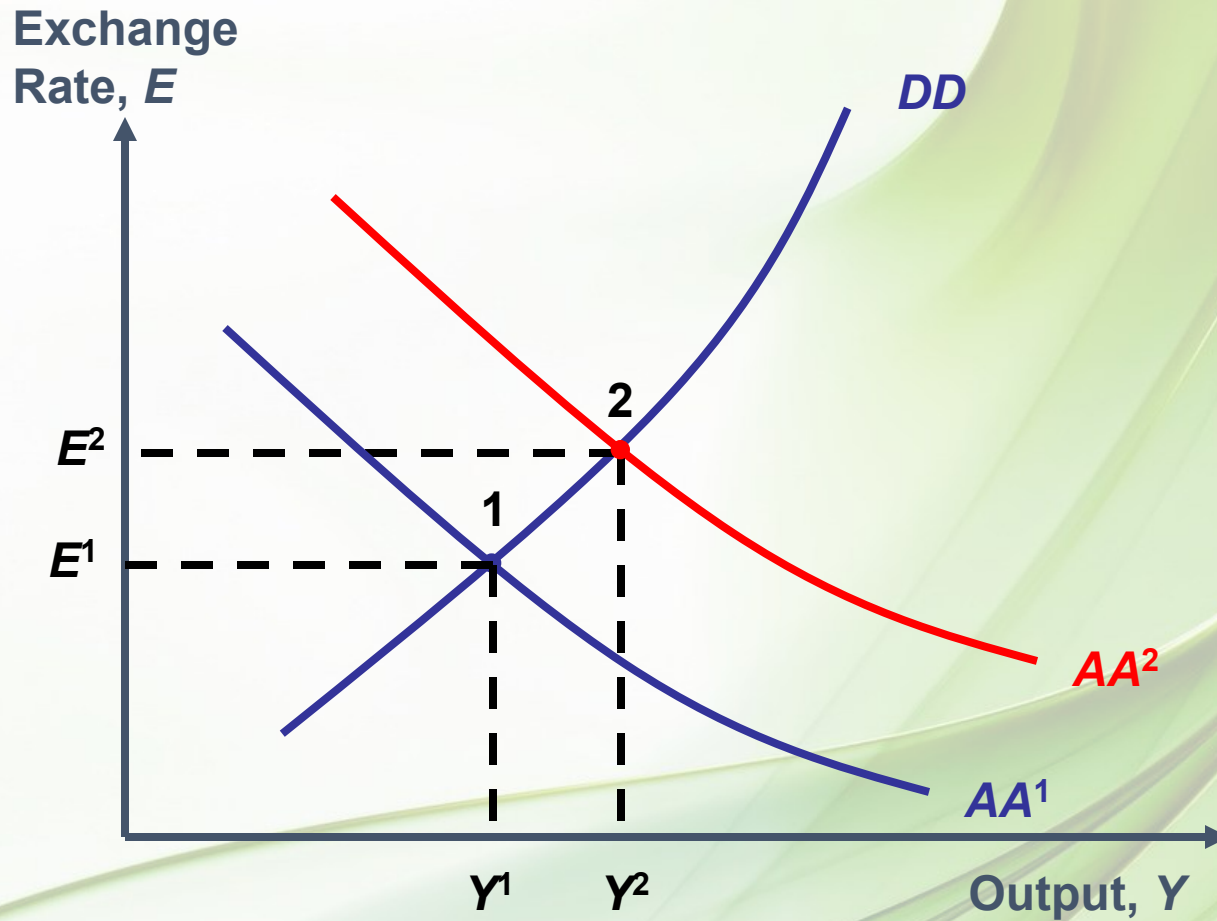


Temporary Changes in Monetary and Fiscal Policy

- Time aspect:
 - **Temporary** policy shifts are those that the public expects to be reversed in the near future and **do not affect the long-run expected exchange rate**.
- Additional assumption
 - **Small open economy**: assume that policy shifts do not influence the foreign interest rate and the foreign price level.

Temporary Changes in Monetary Policy

Effects of a Temporary Increase in the Money Supply



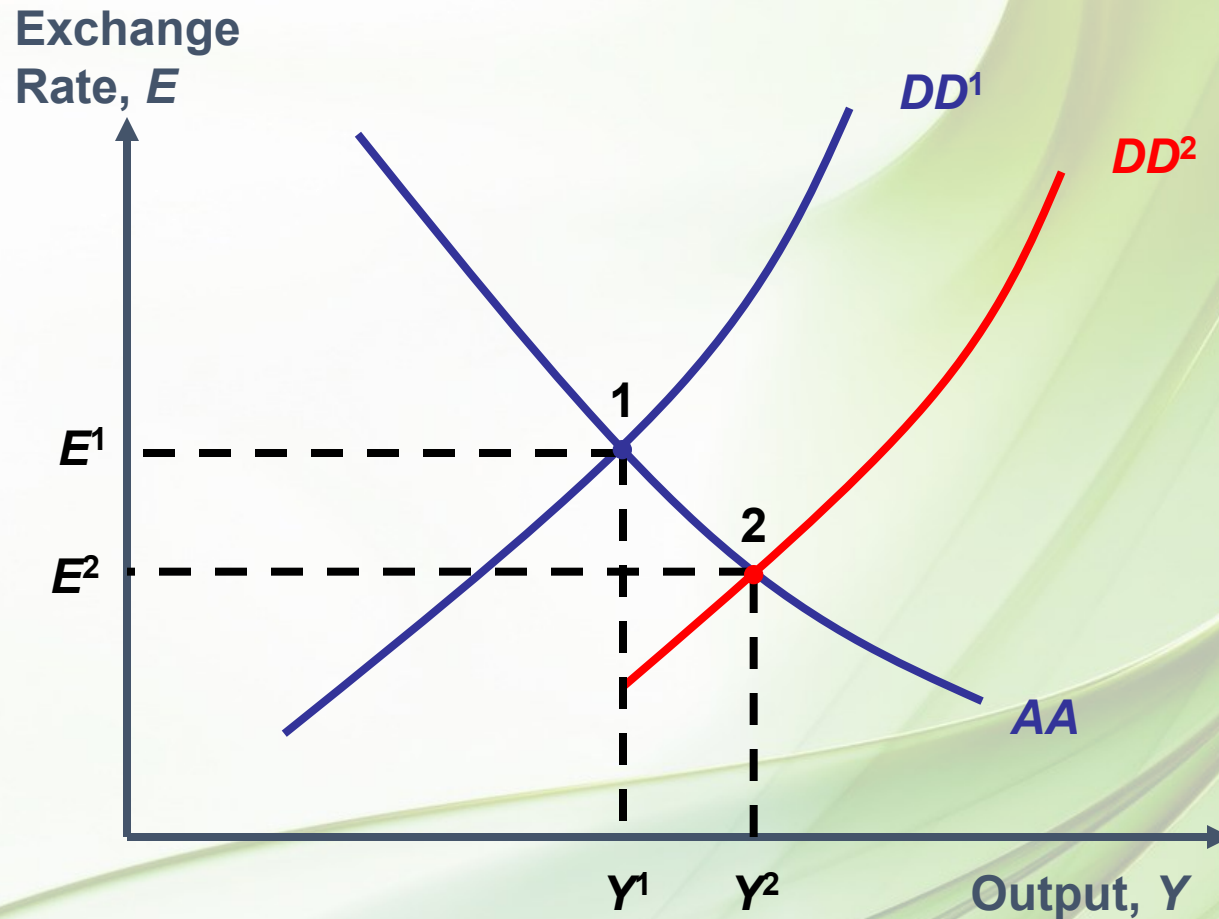
Temporary Changes in Monetary Policy

- **Monetary Policy**

- An increase in money supply (i.e., expansionary monetary policy) raises the economy's output.
 - The increase in money supply creates an excess supply of money, which lowers the home interest rate.
 - As a result, the domestic currency must depreciate (i.e., home products become cheaper relative to foreign products)
 - Domestic products relative to foreign products are cheaper, so that aggregate demand and output increase until a new short-run equilibrium is achieved.

Temporary Changes in Fiscal Policy

Effects of a Temporary Fiscal Expansion



Temporary Changes in Fiscal Policy

- **Fiscal Policy**

- An increase in government spending, a cut in taxes, or some combination of the two (i.e, expansionary fiscal policy) raises output.
 - The **increase in output raises the transactions demand** for real money holdings,
 - Increase in the home interest rate.
 - As a result, the domestic currency must appreciate (E falls)

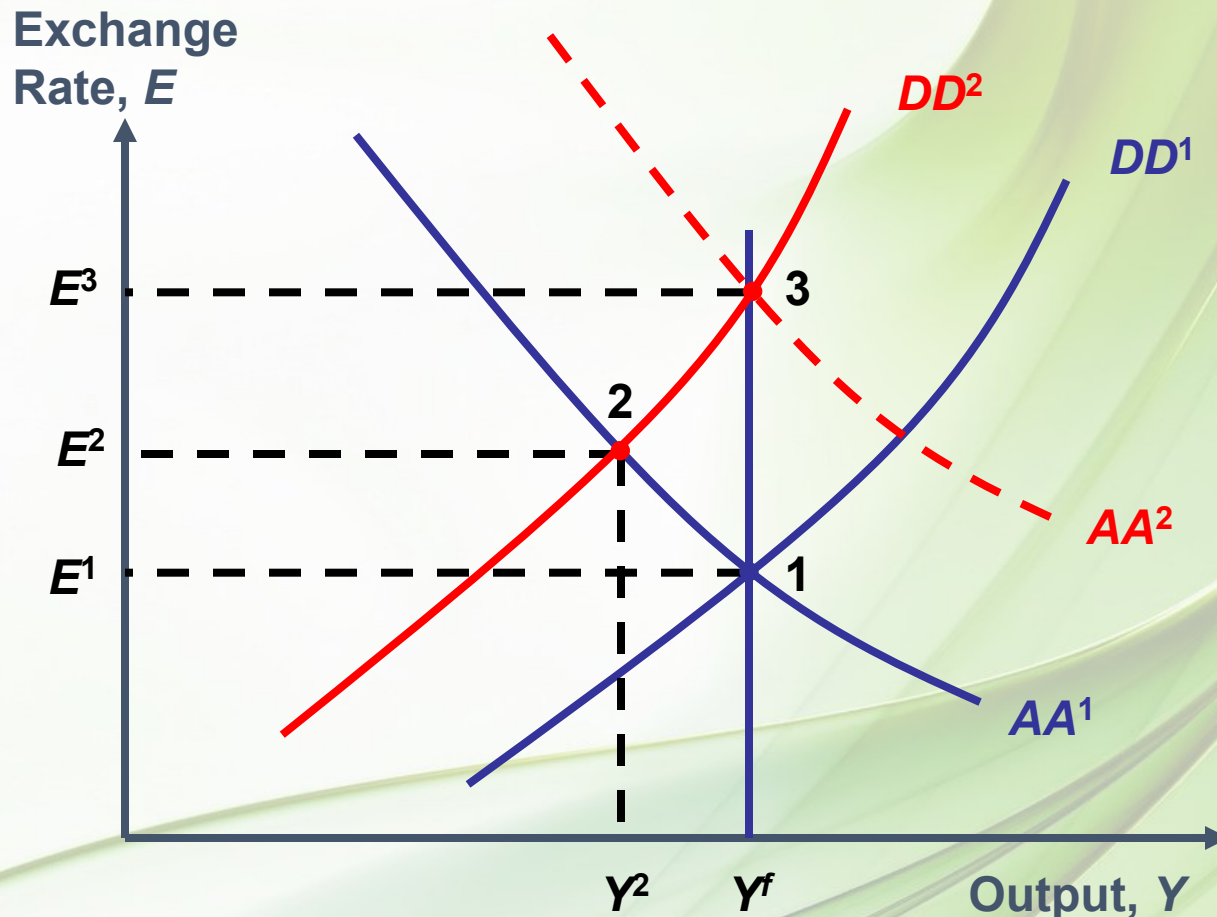
Temporary Changes in Monetary and Fiscal Policy

- **Policies to Maintain Full Employment**

- Temporary disturbances that lead to recession can be offset through expansionary monetary or fiscal policies.
 - Vice versa: temporary disturbances that lead to overemployment can be offset through contractionary monetary or fiscal policies.

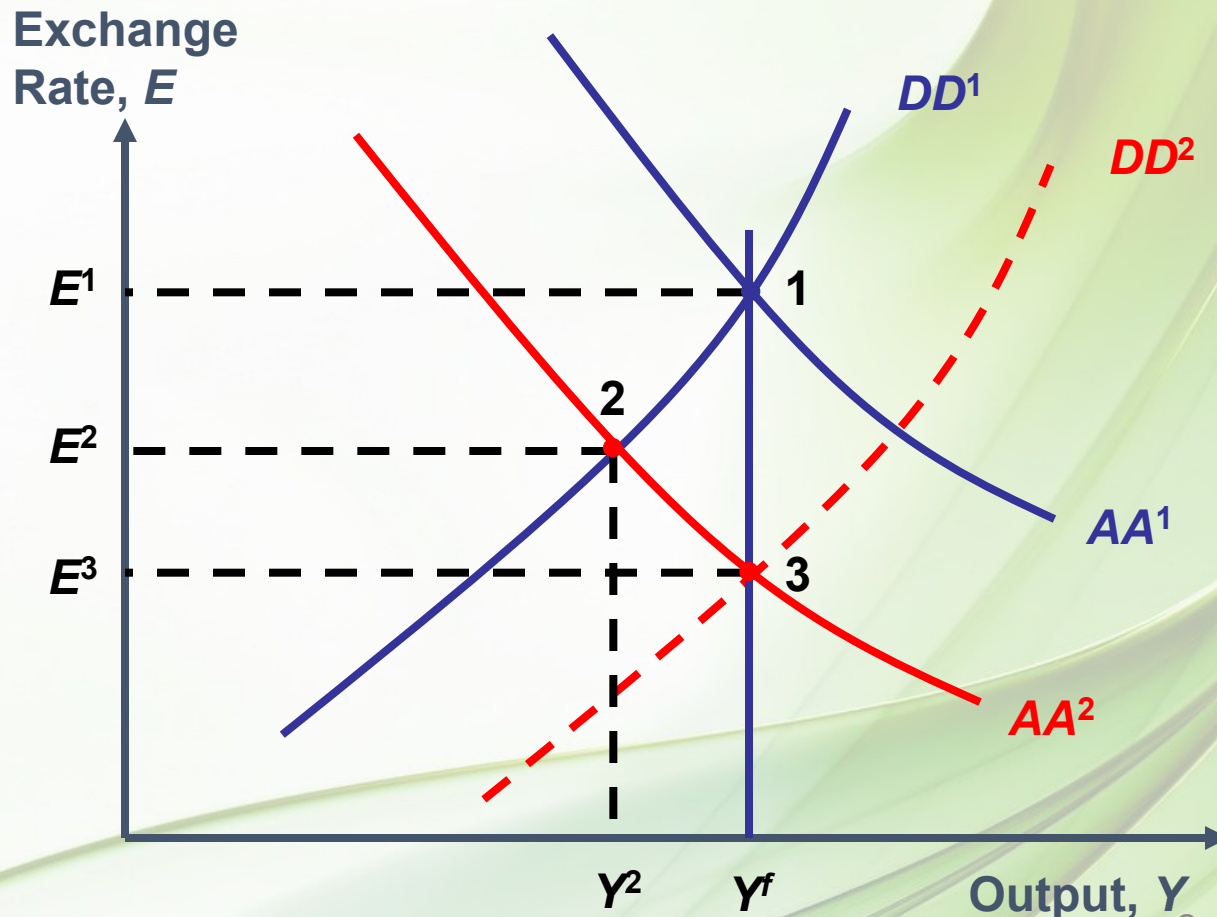
Temporary Changes in Monetary and Fiscal Policy

Maintaining Full Employment After a Temporary Fall in
World Demand for Domestic Products



Temporary Changes in Monetary and Fiscal Policy

Policies to Maintain Full Employment After a Money-Demand Increase

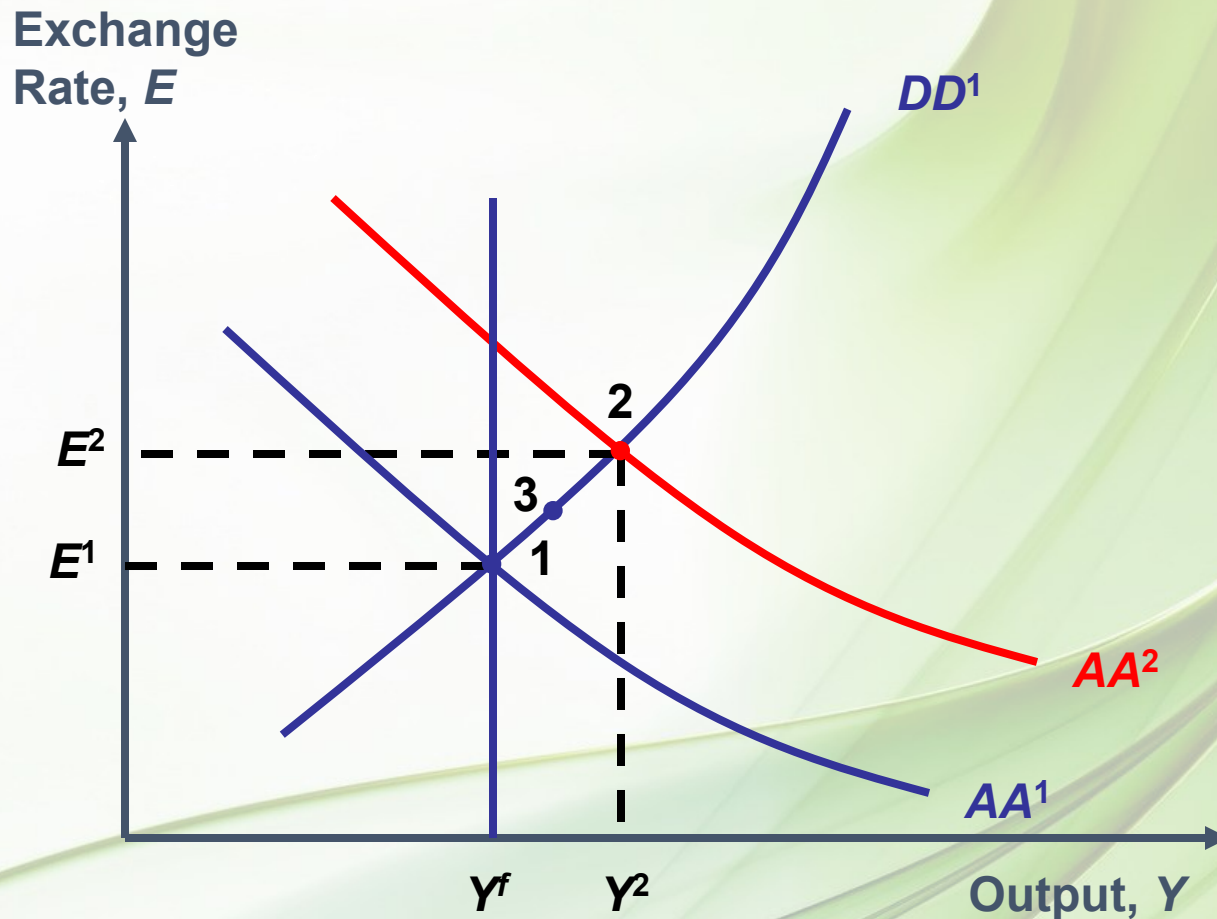


Permanent Shifts in Monetary and Fiscal Policy

- A permanent policy shift affects not only the current value of the government's policy instrument but also the ***long-run exchange rate***.
 - This affects expectations about future exchange rates.
- **A Permanent Increase in the Money Supply**
 - A permanent increase in the money supply causes the expected future exchange rate to rise proportionally.
 - As a result, the **upward shift in the AA schedule is greater** than that caused by an equal, but transitory, increase (compare point 2 with point 3).
 - Why? Think about PPP.

Permanent Shifts in Monetary Policy

Short-Run Effects of a Permanent Increase in the Money Supply

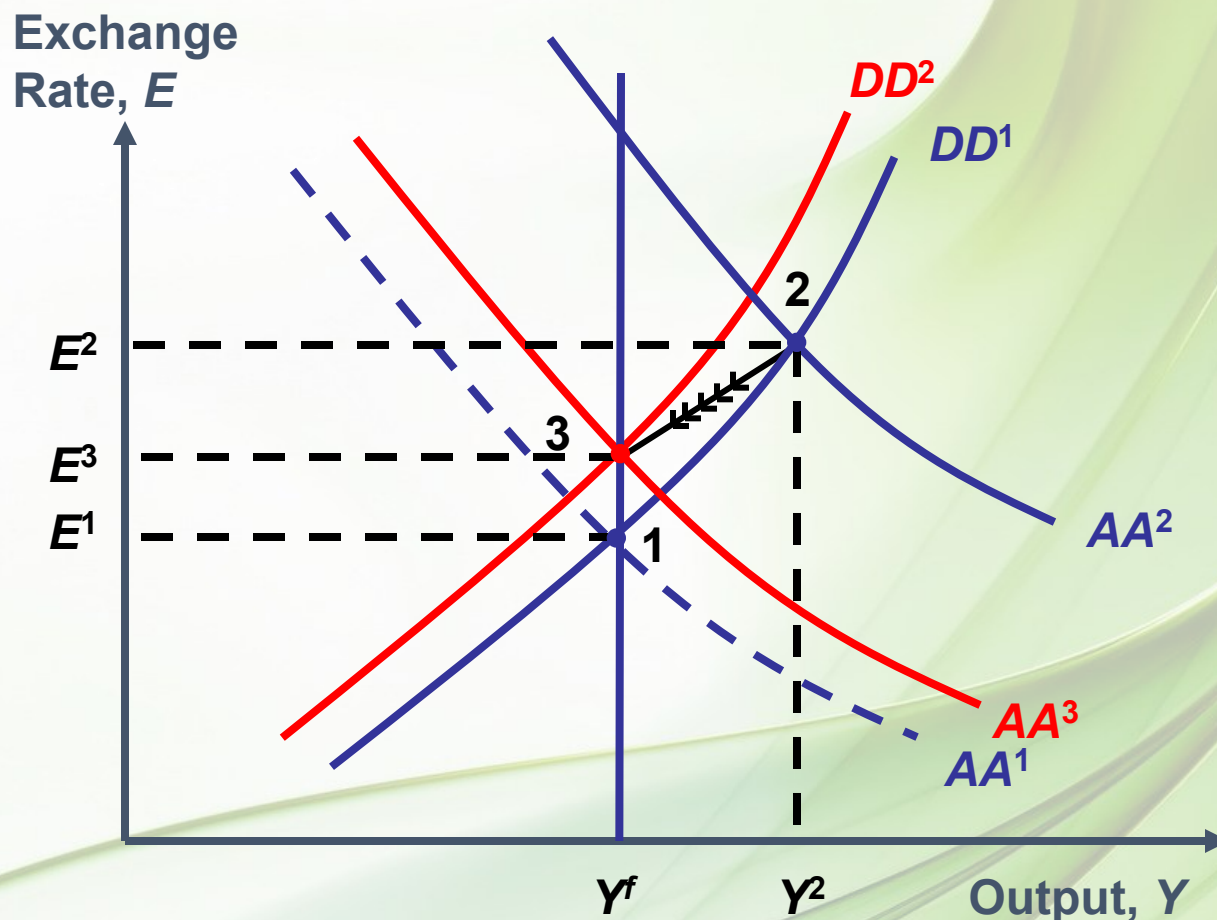


Permanent Shifts in Monetary and Fiscal Policy

- **Adjustment to a Permanent Increase in the Money Supply**
 - The permanent increase in the money supply raises output above its full-employment level.
 - As a result, the price level increases to bring the economy back to full employment.
 - How about the long run: next picture shows the adjustment back to full employment.

Permanent Shifts in Monetary Policy

Long-Run Adjustment to a Permanent Increase in the Money Supply



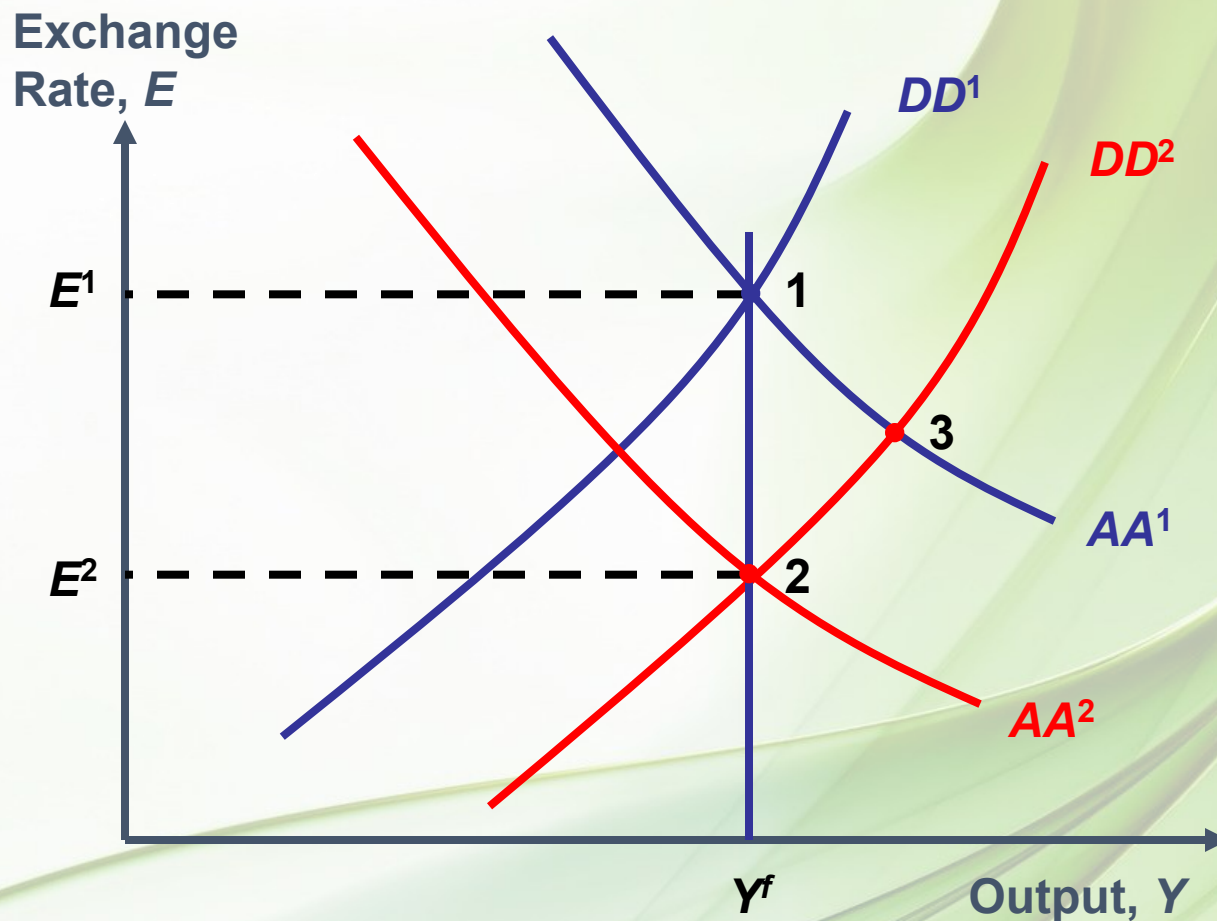
Permanent Shifts in Fiscal Policy

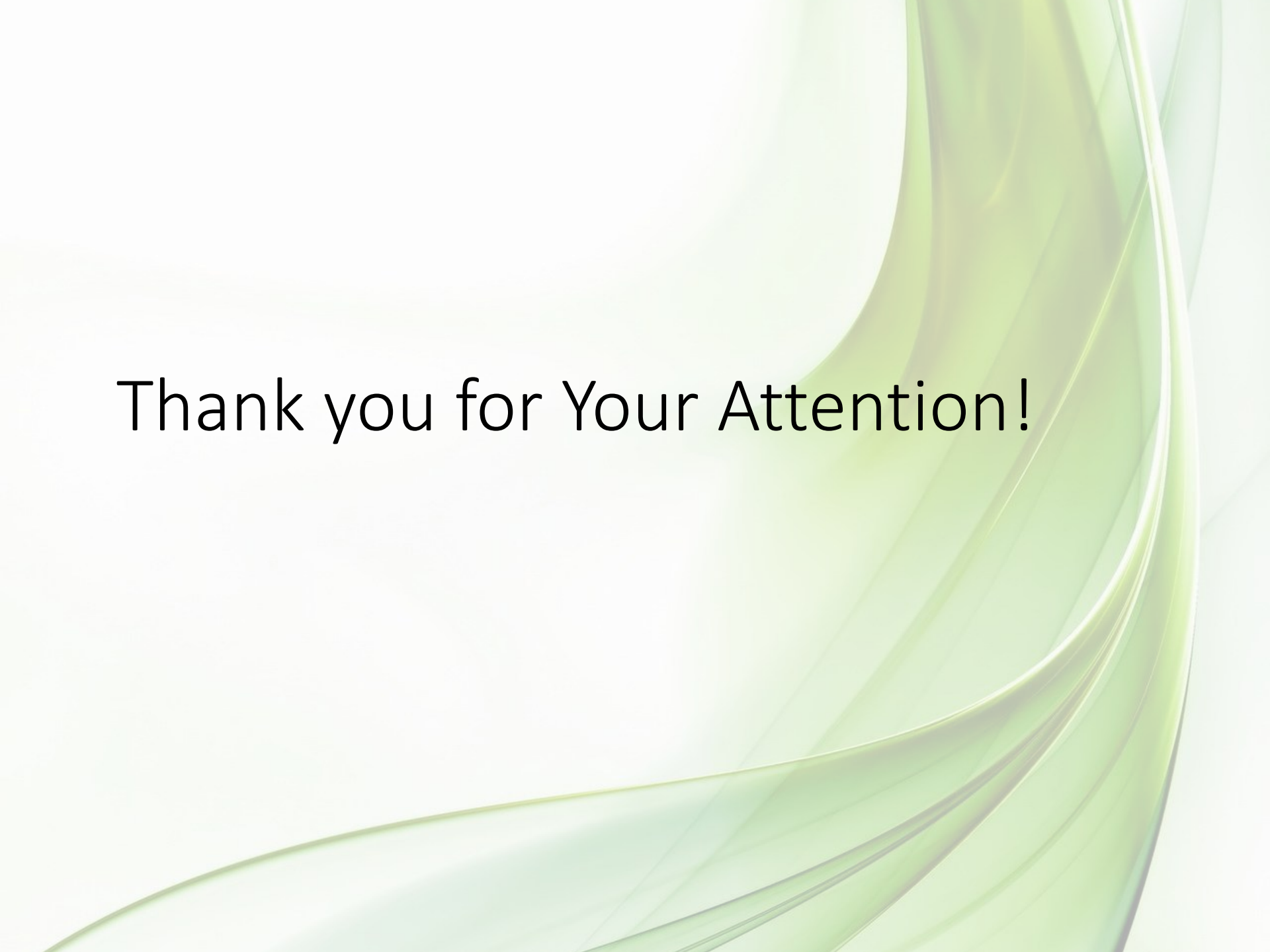
- **A Permanent Fiscal Expansion**

- A permanent fiscal expansion changes the long-run expected exchange rate.
 - If the economy starts at long-run equilibrium, a permanent change in fiscal policy has no effect on output.
 - It causes an immediate and permanent exchange rate jump that offsets exactly the fiscal policy's direct effect on aggregate demand.

Permanent Shifts in Fiscal Policy

Effects of a Permanent Fiscal Expansion



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Thank you for Your Attention!

References

- Cuestas et al (2020): Real exchange rate misalignments in CEECs: Have they hindered growth?
- Maurer (2023): The Divergence of Price Levels in the European Union
- Krugman & Obstfeld & Melitz: International Economics (chapter 17)
- Mao Takongmo (2019): Revisiting the AA-DD Model in Zero Lower Bound
- Baldwin & Krugman (1989): Persistent Trade Effects of Large Exchange Rate Shocks